DI40001 DIICSU Honours Project

Structural Design and Mechanical Simulation of Vertical Roll Bending Forming Equipment for Large-Diameter Thin-Walled Ring Parts

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Abstract

Large-diameter thin-walled stainless steel rings are critical structural components in industries such as aerospace, energy, and petrochemical processing, where high strength-to-weight ratio and dimensional precision are essential. Conventional horizontal roll-bending systems face limitations in handling large, flexible workpieces due to gravity-induced sagging and poor process continuity between uncoiling, forming, and cutting stages. This project addresses these limitations through the structural design and mechanical simulation of a vertical integrated roll-bending system for manufacturing large-diameter thin-walled 304 stainless steel rings (target diameter ϕ12 m, strip thickness 4–10 mm, width 800–2200 mm). A vertical symmetric three-roll layout is adopted and combined with a slide-rail uncoiling unit and a laser cutting module to form a single continuous production line, improving positional stability and process continuity compared with conventional horizontal arrangements.

The complete system was modelled in SolidWorks as a parametric CAD assembly of approximately 31 component bodies. Analytical checks confirmed that the top drive roll ($D_1 = 500$ mm) achieves a mid-span deflection of 0.24 mm/m, within the 0.25 mm/m circularity requirement, while frame buckling and weld fatigue analyses yield safety factors of 13.7 and utilisation ratio 0.70 respectively. Nonlinear finite element simulations were carried out in ANSYS using a bilinear isotropic hardening plasticity model. Static simulations for strip gauges of 4, 6, 8, and 10 mm produced peak Von Mises stresses of 253.6, 298.9, 372.4, and 445.8 MPa, all below the material UTS of 520 MPa. A transient dynamic rolling simulation confirmed bending onset at $t = 0.9$ s and a steady-state mean transverse deflection of 8.3 mm, corresponding to a predicted bending diameter of approximately 10.84 m, within 10% of the 12 m design target. The remaining discrepancy is consistent with expected elastic springback after unloading. These results confirm the mechanical feasibility of the proposed vertical roll-bending architecture and provide a structured basis for subsequent experimental validation.

Keywords: thin-walled ring forming; vertical roll bending; 304 stainless steel; finite element analysis; structural design; springback

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Time moves swiftly, and somehow I have arrived at the moment of writing these words. I am deeply grateful to everyone who has crossed my path along the way — the mentors who offered guidance, the friends who offered company, and the strangers whose kindness left a mark. To my supervisor and colleagues at DIICSU, thank you for your patience and support throughout this project. Special thanks go to my parents, whose unwavering support and encouragement have been my greatest source of strength throughout this journey.

The sun always rises again. So does life. I hope that every step I take is one I will never regret, and that as long as I hold a bearing, no matter how small I may feel, I will never truly lose my way.

1 Introduction

1.1 Background and Motivation

Thin-walled ring components are generally defined as ring parts whose wall thickness-to-inner-radius ratio is less than 1:20, characterised by small wall thickness and high structural strength. With continued advances in modern engineering, such components are extensively used across a broad range of industries including electromechanical equipment, automotive manufacturing, rail transit, nuclear power, petrochemical processing, and coal liquefaction. Representative applications include wind turbine towers (both offshore and mountain-top installations), aircraft fuselages and wing structures, nuclear reactor pressure vessels, turbine casings, boilers, and pressure vessels, as illustrated in Figure 1.



(a) (b)

Figure 1: Applications of thin-walled ring components in engineering [1].

In aerospace applications in particular, thin-walled ring parts often serve as critical load-bearing structures. Components such as heavy launch vehicle airframes and crewed lunar lander bodies must withstand extreme flight conditions including high-manoeuvre overloads and the high temperature and pressure encountered during atmospheric re-entry. Consequently, aerospace thin-walled components must satisfy not only strict dimensional and geometric accuracy requirements, but also exceptionally high load-bearing capacity and structural reliability. Furthermore, because the wall thickness is extremely small relative to the radial dimensions, these parts are highly susceptible to deformation during processing, making it difficult to meet precision requirements and severely affecting product quality [2]. In addition, such ring and shell parts are increasingly trending towards larger scale, monolithic construction, lightweight design, and higher reliability, placing ever greater demands on manufacturing processes.

The target workpiece in this project is a large-diameter thin-walled ring manufactured from 304 stainless steel. This alloy is widely employed in structural applications owing to its corrosion resistance and moderate formability. Nevertheless, its forming behaviour presents significant process control challenges. Springback is a particularly critical issue, and material anisotropy has been shown to influence final geometry upon unloading [3]. Consequently, the design of forming equipment must address not only the geometric layout of the machine, but also the nonlinear mechanical response of the workpiece

material.

Roll bending is an established method for producing large curved plates, shells, and ring-like components. Analytical investigations of three-roller cylindrical bending have demonstrated that the forming response is governed by the coupled effects of roll geometry, plate thickness, and material properties [4]. Concurrent developments in equipment design indicate that roll-bending machines are advancing towards higher positional precision and enhanced automatic control capability [5]. Dedicated forming technologies for thin-walled cylindrical rings have also been reported, underscoring the engineering significance of this manufacturing challenge [6].

1.2 Aim and Objectives

The specific problem addressed in this project is how to design a vertical roll-bending system for manufacturing large-diameter thin-walled 304 stainless-steel rings. Very large and thin workpieces are difficult to handle and form accurately, and bending quality is sensitive to material behaviour and process parameters [2]–[4]. Existing studies have separately discussed thin-walled component manufacturing, ring-forming methods, bending mechanics, springback behaviour, and precision roll-bending equipment [2]–[6], but limited work has combined these aspects in one design framework for a vertical integrated three-roll system aimed at approximately $\phi 12\,\mathrm{m}$ thin-walled 304 stainless-steel rings. This project therefore focuses on linking equipment layout, structural design, and mechanical simulation for this specific application.

The overall aim of this project is to design and evaluate an integrated vertical roll-bending system for manufacturing large-diameter thin-walled 304 stainless-steel rings.

To achieve this aim, the project has the following objectives:

(i) analyse the forming requirements of large-diameter thin-walled rings and justify the use of a vertical symmetric three-roll layout;

(ii) determine the main structural and geometric parameters of the proposed equipment;

(iii) build a complete three-dimensional model of the system in SolidWorks;

(iv) establish a nonlinear finite element model in ANSYS to study deformation, stress, and contact behaviour during the roll-bending process; and

(v) use the simulation results to assess the feasibility of the design and support later optimisation.

The report as a whole includes the content shown in the Figure 2.

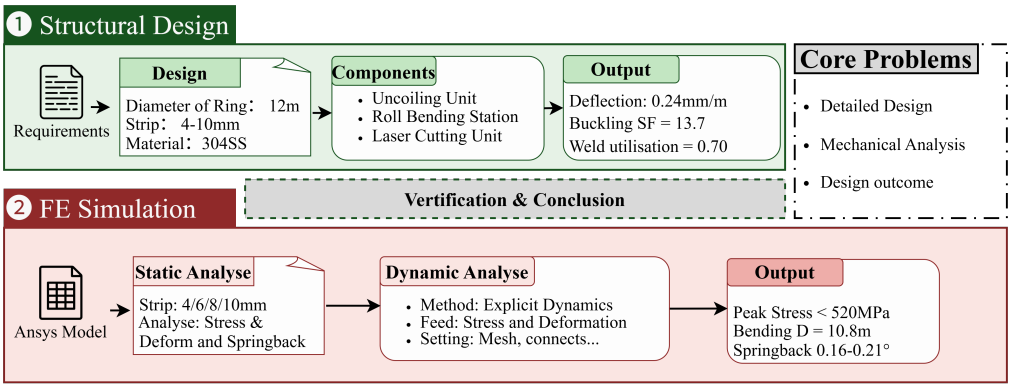


Figure 2: Overview of the thesis structure.

1.3 Scope and Limitations

This project focuses on the structural design and mechanical simulation of the proposed equipment system. The work includes concept selection, parameter design, three-dimensional modelling, and nonlinear finite element analysis of the roll-bending process. It does not include prototype manufacturing, full-scale experimental testing, or the full design of the electrical and hydraulic control system.

These boundaries are consistent with the scope of an honours project and reflect practical constraints in time, resources, and access to physical testing facilities. The principal contribution of this study is therefore the development and rigorous evaluation of an engineering design concept, rather than the delivery of a fully validated industrial prototype.

1.4 Report Outline

The remainder of this report is organised as follows.

(i) Section 2 reviews rolling-based forming processes, ring-related manufacturing technologies, and the theoretical basis of springback and finite element analysis in roll bending;

(ii) Section 3 presents the design specifications, the geometric and mechanical basis of the three-roll configuration, and the parameter design of all principal subsystems;

(iii) Section 4 describes the CAD realisation of the integrated system in SolidWorks, covering the main bending station, uncoiling unit, laser cutting module, and base platform;

(iv) Section 5 presents and discusses the ANSYS simulation results, including static stress and deformation for four strip thicknesses, dynamic rolling analysis, and target curvature verification; and

(v) Section 6 summarises the main findings, states the principal contributions, and identifies directions for future experimental validation and design refinement.

2 Literature Review

2.1 Background

Rolling-based forming belongs to a broader family of incremental deformation processes developed to achieve material-efficient shape generation under demanding geometric and mechanical constraints. Existing reviews show that this family includes multiple process branches, such as flat rolling, shape rolling, and ring-related rolling processes, each characterised by distinct application ranges, deformation modes, and process limitations [7]. Within this broader landscape, ring-related rolling and roll-bending routes are particularly relevant to the manufacture of large annular and cylindrical parts, where conventional subtractive or segmented fabrication methods often suffer from poor material utilisation, reduced geometric consistency, or high downstream assembly burden.

The relevance of this topic has increased as modern engineering components have become larger, thinner-walled, and more integrated. Recent studies on extreme-size integrated metal components and thin-walled cylindrical rings with complex geometric features indicate that manufacturing challenges are no longer restricted to whether a target shape can be formed, but increasingly concern dimensional stability, structural integrity, process coordination, and controllable deformation over large and low-stiffness workpieces [8], [9]. These requirements are especially acute in aerospace, energy, and other high-end manufacturing settings, where thin-walled ring-like parts must satisfy both lightweight and high-reliability demands.

To position the present project within this broader body of knowledge, Table 1 summarises the main rolling-based forming branches discussed in the literature and highlights why ring-related rolling remains a distinct and demanding process category.

The historical development of ring-related rolling technologies shows that the field has evolved through more than changes in machine arrangement alone. A historical survey of ring rolling technology indicates that early development focused primarily on the feasibility of near-net-shape production and the expansion of industrial capability, whereas later work increasingly addressed process behaviour, production equipment, investigation methods, and operational improvement [10]. This pattern shows that technological maturity has depended not only on stronger machines, but also on improved understanding of deformation behaviour and more reliable control of process conditions.

More recent reviews suggest that the roll-forming and roll-bending field has continued to evolve toward higher precision, more complex geometries, and stronger integration with digital manufacturing approaches [11]. In practical terms, this evolution is reflected in the transition from conventional roller configurations toward multi-roll and CNC-assisted systems intended to improve positioning accuracy, repeatability, and forming flexibility. Jing et al. reported a high-precision four-roll CNC bending machine with an automatic control model, demonstrating the feasibility of integrating servo-assisted actuation into roll-bending equipment [5].

Table 1: Representative rolling-based incremental deformation processes [7].

Process type	**Typical products**	**Main strength**	**Relevance to this project**
Flat rolling	Sheets and plates	High productivity	Material-preparation route, insufficient for large annular geometry
Shape rolling	Profiles and sections	Efficient cross-section generation	Relevant to deformation-path design, less suited to large rings
Ring rolling	Rings, annular parts	Near-net-shape forming	Closest process family to the present work
Roll bending	Curved plates, shells	Flexible curvature generation	Directly relevant to the vertical three-roll design
Other processes	Process-specific	Expanded capability	Useful for comparison only

However, the literature also implies an important engineering trade-off. Although more sophisticated machine architectures can improve controllability and automation, they also tend to increase system complexity, coordination demands, and implementation cost. For applications involving large-diameter thin-walled rings, the more relevant question is how to balance structural simplicity, forming capability, positional stability, and subsequent validation, rather than how to maximise equipment sophistication in isolation.

2.2 Critical Synthesis

At the theoretical level, roll-bending and ring-related rolling processes are governed by the coupled effects of contact, bending deformation, plastic flow, and elastic recovery after unloading. Process quality depends not only on machine configuration, but also on how force transmission, geometric compatibility, and springback are understood and controlled. Recent analytical and finite element work on the three-roller cylindrical bending process shows that rolling force is strongly influenced by plate thickness, roller diameter, and material yield strength, and that these variables are mechanically coupled rather than independently tunable design parameters [4]. This is directly relevant to the present project, where roller dimensions, centre distance, and the target bending radius must be selected under structural and process constraints simultaneously.

Among the theoretical issues, springback remains one of the most persistent challenges. Classical simulation studies have shown that reliable springback prediction depends sensitively on constitutive assumptions, through-thickness stress distribution, and

contact treatment in numerical analysis [12], [13]. This is especially critical for thin-walled parts, where even modest curvature recovery may produce substantial dimensional error when the target diameter is large. Accordingly, springback should not be treated as a minor post-process deviation, but as a central theoretical problem in precision rolling-based forming.

The literature also demonstrates a clear methodological shift from simplified analytical interpretation toward simulation-assisted understanding. Dynamic finite element studies have shown that roll-bending processes are more appropriately analysed as nonlinear, contact-dominated deformation systems in which stress evolution, deformation history, and final geometry must be considered together [14]. Specific work on 304 stainless steel confirms that residual stress and springback behaviour are materially sensitive and require more than empirical estimation when high precision is needed [15]. Related research on precision bending strategies further suggests that analytical modelling and data-supported correction methods can improve final accuracy [16].

Research on bending-angle distribution and springback mitigation indicates that final forming accuracy depends strongly on parameter scheduling, boundary conditions, and multi-stage optimisation rather than solely on machine type [17], [18]. This implies that structural design, process mechanics, and parameter optimisation should be treated as a connected problem rather than as isolated topics.

2.3 Identification of the Gap

Taken together, the literature demonstrates substantial progress in four areas: the broader understanding of rolling-based forming routes, the historical development of ring-related rolling technologies, the evolution of production equipment, and the theoretical and numerical analysis of force transmission, springback, and process optimisation. Nevertheless, these research strands remain only partially integrated for the engineering case considered in this project.

Existing studies have either focused on broad process families and industrial context [7], [8], on specialised ring-forming geometries and development trajectories [9]–[11], on advanced high-precision machine systems [5], or on force analysis, springback prediction, and simulation-assisted optimisation under particular forming conditions [4], [12]–[18]. Far less attention has been given to how these perspectives can be integrated into a unified design-and-validation route for manufacturing large-diameter thin-walled stainless-steel rings using a vertical three-roll system with coordinated upstream and downstream units.

The contribution of the present project therefore lies in addressing the insufficiently connected state of the literature by linking equipment architecture selection, mechanically grounded parameter design, and finite-element-based validation for a vertical three-roll integrated manufacturing system aimed at large thin-walled ring components.

3 Methodology

3.1 Design specifications and overall scheme selection

The present project focuses on the design of an integrated vertical roll-bending system for manufacturing large thin-walled cylindrical rings from 304 stainless steel sheet. According to the project requirements, the target product diameter is 12 m, which corresponds to a target forming radius of

$$R = \frac{D}{2} = \frac{12000}{2} = 6000\,\text{mm}. \tag{1}$$

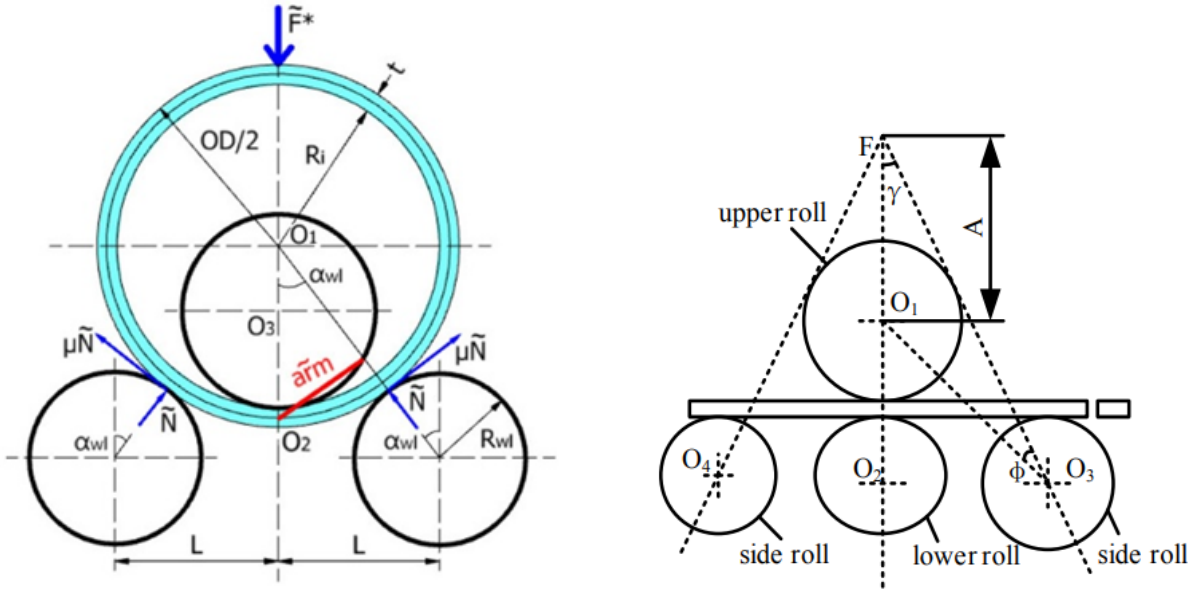
The equipment is required to process sheet thicknesses from 4 mm to 10 mm and widths from 800 mm to 2200 mm. In addition to the bending function itself, the production route must allow coil feeding, forming, and final cutting within one coordinated layout.

The overall design therefore started from three constraints. The workpiece is thin relative to its final diameter, so positional stability during forming is a major concern. In addition, the equipment must provide sufficient bending capacity for the maximum thickness case. The layout must also remain structurally simple enough for engineering implementation and later simulation-based verification.

Different roll configurations were considered at the concept generation stage. A two-roll system was not selected because, although it can bend sheet metal, its deformation path and support condition are less suitable for the present large-diameter heavy-duty application, where stable control of the bending path and load transfer is required. A four-roll machine provides strong forming capability and good control in many industrial applications, but its structure is more complex, its manufacturing and control requirements are higher, and it is less economical for the present design target. By contrast, a symmetric three-roll configuration provides a clearer force path and a simpler mechanical structure while still allowing controllable bending of large plates. Previous work on symmetrical three-roller setting round processes has also shown that the static bending stage can be described through a mechanical model, which supports the use of a three-roll configuration as a rational analytical design basis [19]. In addition, studies on three-roll bending machine tools further indicate that geometric design and displacement coordination are central to the final forming performance [20].

Figure 3 illustrates the working principles of the three-roll and four-roll configurations considered during concept selection.

The project further adopted a vertical rather than a conventional horizontal layout. Earlier research on vertical-roll forming processes showed that vertical roll arrangements can improve forming characteristics and offer a distinct process route relative to conventional layouts [21]. For the present study, the vertical arrangement was chosen because it is more compatible with the intended integrated route from coil feeding to ring forming and cutting. On this basis, the final concept was organised into three main subsystems: the central three-roll forming and support subsystem, the uncoiling subsystem, and the auxiliary cutting subsystem.



(a) Symmetric three-roll bending [4] **(b)** Four-roll bending [22]

Figure 3: Working principles of three-roll and four-roll bending configurations.

3.2 Design of the central three-roll forming subsystem

3.2.1 Functional role of the central subsystem

The central subsystem is the core unit of the whole machine. Its main task is to impose the required plastic curvature on the sheet so that the final rolled plate approaches the target cylindrical diameter of 12 m. In the adopted scheme, the upper roll acts as the main driving roll, while the two lower rolls provide support and bending displacement in a symmetric manner. The support platform around the roll set is designed to maintain the stability of the large sheet during forming and to reduce positioning error during the transition from feeding to bending.

3.2.2 Geometric basis of three-roll bending

The key design problem of the three-roll subsystem is to determine the geometric relationship between the roll positions and the required final curvature. The target cylinder radius is

$$R = 6000\,\text{mm}. \tag{2}$$

For a symmetric three-roll arrangement, if the half-spacing of the two lower rolls is denoted by a, then the full lower-roll centre distance is

$$L = 2a. \tag{3}$$

Under a simplified geometric bending assumption, the equivalent displacement required to impose a target radius can be approximated by the sagitta relation

$$\delta = R - \sqrt{R^2 - a^2} \tag{4}$$

where δ is the effective bending displacement associated with the roll geometry. This relation shows directly that, for a fixed target radius R, the required displacement depends on the adopted lower-roll spacing. The geometric relationship between roller position and final curvature is also consistent with previous analytical and empirical studies of three-roller cylindrical bending, where the final radius is governed by the roller position and contact geometry [23]. Likewise, analytical–numerical studies of three-roller systems have shown that the bending radius is directly linked to the vertical displacement imposed by the roll arrangement [24].

In the present design, the lower-roll centre distance was finally taken as

$$L = 600\,\text{mm}, \qquad a = 300\,\text{mm}. \tag{5}$$

Substituting these values into Eq. (4) gives

$$\delta = 6000 - \sqrt{6000^2 - 300^2} \approx 7.50\,\text{mm}. \tag{6}$$

This value is small compared with the final radius, which is consistent with the fact that a large-diameter thin-walled ring requires a relatively gentle but precisely controlled bending path.

This geometric result also helps explain why the roll spacing cannot be chosen arbitrarily. If the lower-roll spacing is too small, the effective bending zone becomes too concentrated, which increases local curvature and reduces the smoothness of the forming path. If the spacing is too large, the required displacement and force rise, and the structural demand on the machine increases. For this reason, the roll spacing was not selected only from structural convenience; instead, it was chosen as a compromise between curvature control, machine load, and layout feasibility. Analytical work on rolling force in three-roller bending further shows that plate thickness, roll diameter, and material yield strength strongly influence the force demand, so the geometric choice must be coordinated with the mechanical design rather than treated independently [4].

3.2.3 Mechanical analysis of the bending load

After the geometric target was defined, the next step was to estimate the mechanical load required for bending. The sheet material was specified as AISI 304 stainless steel, with Young's modulus taken as 1.93×10^5 MPa, Poisson's ratio as 0.3, and yield strength as 205 MPa, consistent with published values for AISI 304 stainless steel [25]. These values were also used in the later ANSYS model.

For a rectangular sheet section of width b and thickness t, the elastic section modulus is

$$W = \frac{bt^2}{6}. \tag{7}$$

If the onset of yielding is used as the initial criterion for bending design, the corresponding bending moment can be written as

$$M = \sigma_y W = \sigma_y \frac{bt^2}{6}, \tag{8}$$

where σ_y is the yield strength of the plate material. Equation (8) shows directly that the bending moment increases linearly with the sheet width and with the square of the thickness. This is one of the main reasons why the 10 mm thickness case governs the structural design.

Taking the maximum design width $b = 2200\,\text{mm}$, the limiting thickness $t = 10\,\text{mm}$, and $\sigma_y = 205\,\text{MPa}$, the section modulus evaluates to $W = 2200 \times 10^2/6 \approx 36{,}667\,\text{mm}^3$, and the corresponding yielding moment is $M = 205 \times 36{,}667 \approx 7.52 \times 10^6\,\text{N·mm}$. This value provides the initial load scale for the roll design. Although the real roll-bending process is elastic–plastic and involves contact effects, it remains useful as a first structural estimate. Previous analytical and finite element studies of three-roller bending similarly use an initial mechanical calculation as the basis for later force refinement [4], [24].

3.2.4 Selection of roll diameters and effective length

The roll diameters were then selected from the combined requirements of load-bearing capacity, curvature control, and structural manufacturability. In general, larger roll diameters reduce local stress concentration and provide a smoother bending path, but they also increase the machine size, mass, and actuation demand. Analytical and numerical studies on three-roller bending have shown that the rolling force is sensitive to upper roll diameter and plate thickness, which confirms that roll diameter is not merely a geometric parameter but also a mechanical one [4].

The present design finally adopted an upper roll diameter of $D_1 = 500\,\text{mm}$ and lower roll diameters of $D_2 = D_3 = 450\,\text{mm}$. The upper roll acts as the main driving roll and therefore requires a larger diameter to ensure adequate stiffness and stable torque transmission. Slightly smaller lower rolls help maintain a compact layout while still providing sufficient support and bending action. This diameter combination also remains compatible with the target centre distance and the required working envelope. The selected values ($L = 600\,\text{mm}$, $D_1 = 500\,\text{mm}$, $D_2 = D_3 = 450\,\text{mm}$, $l_{\text{eff}} = 2200\,\text{mm}$) were therefore not assigned arbitrarily; they reflect the combined geometric, mechanical, and structural constraints of the design. Research on symmetrical three-roller processes confirms that the static bending stage is governed by a coupled geometric–mechanical relation rather than a single empirical dimension [19], and recent work on three-roll bending machines has emphasised that displacement coordination and structure must be designed together [20].

The effective roll length must exceed the maximum sheet width while allowing for lateral tolerance and support conditions. Since the design width range is 800 to 2200 mm, the effective roll length was set to $l_{\text{eff}} = 2200\,\text{mm}$ to accommodate the maximum width case with a practical structural margin.

3.2.5 Drive and support structure

The drive route of the central subsystem was designed around hydraulic actuation. The upper roll is driven in rotation to pull the sheet through the roll gap, while the lower rolls provide the bending displacement required for curvature generation. This arrangement follows the general design logic of three-roll systems, in which the forming radius is controlled through the coordinated displacement of the roll set [23], [24]. The support platform beneath and around the roll unit was designed as an integral structural element rather than a separate accessory, because the target sheet is large and prone to instability during handling and forming. Its function is to support the sheet path, reduce unwanted deflection, and maintain alignment during the forming sequence.

3.3 Structural design of the uncoiling subsystem

The raw material enters the process in coil form, so an uncoiling subsystem is required upstream of the three-roll unit. Its role is to provide stable material feeding, maintain the strip path, and reduce handling operations before the plate enters the forming region. A vertical uncoiling arrangement was selected to match the overall vertical route of the proposed equipment.

The uncoiling unit is built around a circular frame on which three rollers are mounted at equal angular spacing. Each roller can travel radially along a slide rail on the frame, allowing the assembly to grip coils of varying inner diameter. The rollers carry rotating wheels on their contact faces; these wheels clamp the coil and allow the strip to be unwound smoothly with low friction. A central vertical column provides structural support for the entire frame. One column on the outer ring is fitted with a pivoting rectangular guide plate that can be rotated to set the strip exit direction, ensuring correct alignment with the bending station entrance. In this sense, the uncoiling subsystem is not an independent machine, but a process-matched feeding unit designed around the requirements of Section 3.2.

3.4 Auxiliary cutting and integrated layout

After bending, the formed strip must be separated to complete the ring-making route. An auxiliary laser cutting subsystem was therefore included at the downstream side of the equipment. Its design focused on positional integration with the main machine rather than on detailed cutting-process optimisation.

The cutting unit is mounted on a sliding base that allows the entire module to traverse along the strip direction. Both the laser emitter head and the receiving end are independently movable along this base, so the cutting span can be adjusted to match the strip width. The laser emitter is additionally equipped with a lead-screw mechanism that drives the emitter head vertically, enabling the laser beam to traverse the full cross-section of the strip and complete the cut. Together with the uncoiling unit and the central roll-bending unit, this arrangement forms a continuous process route from coil feeding to final ring separation.

3.5 Analytical methods and verification route

The design methodology followed a structured sequence: design specification, concept generation and selection, analytical calculation, CAD realisation, and finite element verification.

The analytical methods used include the sagitta-based geometric relation (Eq. 4) for roll geometry sizing, the elastic section modulus formula for bending load estimation, and analytical checks for roll deflection, frame buckling, and weld fatigue. These calculations provided a mechanically grounded basis for selecting all principal structural dimensions before any numerical model was built. Previous three-roller studies have likewise combined analytical calculation with finite element simulation when estimating bending force and final curvature [4], [24].

Following analytical design, the equipment was implemented as a three-dimensional SolidWorks assembly. This step served as a geometric verification: it confirmed that the analytically determined dimensions were spatially compatible and could be realised as a coherent mechanical assembly. The CAD model also provided the geometry for later finite element analysis.

The finite element verification stage used the same core design inputs as the analytical stage, including roll diameters, centre distance, material properties, and target radius. Static simulations were carried out for four strip thicknesses to examine stress and deformation under representative forming conditions, and a transient dynamic simulation was used to assess steady-state bending response and target curvature achievement. The detailed modelling assumptions, mesh strategy, boundary conditions, and result interpretation are presented in the subsequent chapter. The purpose of the present section is to define how the machine was designed; the following chapters evaluate whether the design satisfies the intended forming task.

4 Implementation

This section presents the detailed implementation of the proposed integrated vertical roll-bending system. The emphasis is on the engineering realisation of the design rather than theoretical derivation. Specifically, this section describes the overall system layout, the CAD assembly of the main and auxiliary modules, the implementation of the vertical three-roll bending station, and the assembly checks carried out to ensure structural and operational consistency. Detailed engineering drawings are provided in the appendix. The present section focuses on the main structural configuration, functional coordination, and assembly-level design.

4.1 Overall System Implementation

The equipment was implemented as an integrated manufacturing system for large-diameter thin-walled stainless-steel rings. The system consists of four main modules: a vertical uncoiling unit, a vertical three-roll bending station, a cutting unit, and a base platform. Together, these modules support a continuous process from coil feeding to strip forming and final cutting.

Figure 4 shows the overall three-dimensional CAD model of the integrated system. The strip is first released from the vertical uncoiling unit, guided into the roll-bending station where the required curvature is generated, and then enters the cutting region for final separation. Detailed engineering drawings are provided in Appendix G.

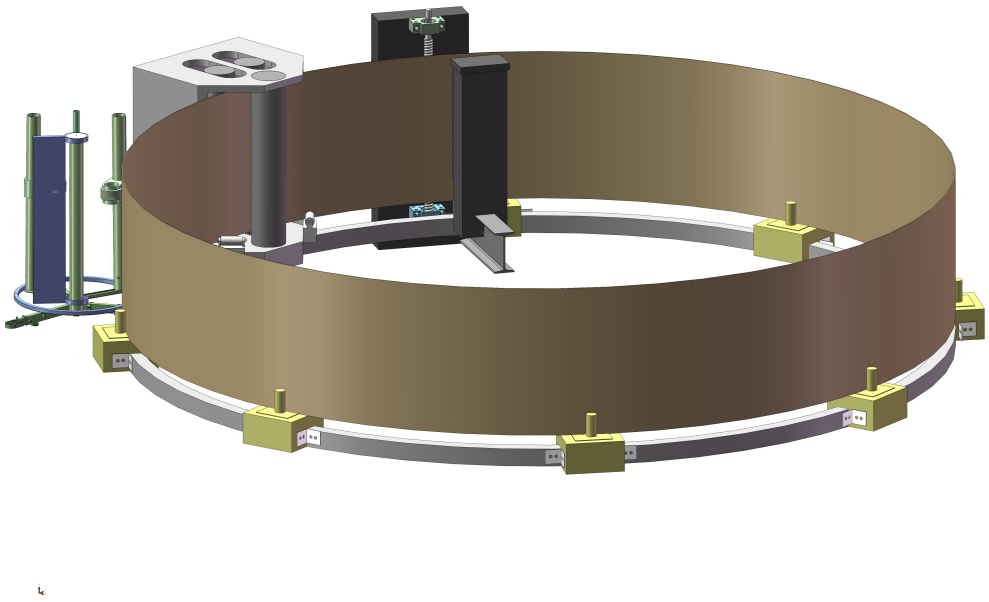


Figure 4: Overall 3D CAD model of the integrated vertical roll bending system.

The implemented system was designed to satisfy the principal project requirements. These requirements are summarised in Table 2. The equipment is intended for 304 stainless-steel strips with a thickness of 4–10 mm and a width of 800–2200 mm, and it is designed to form a ring with a target diameter of 12 m. Hydraulic actuation is used as the main power source for roll motion and auxiliary mechanical actions.

Table 2: Main implementation targets of the integrated system

Item	Target range or value
Strip material	304 stainless steel
Strip thickness	4 to 10 mm
Strip width	800 to 2200 mm
Target formed diameter	12 m
Main forming structure	Vertical symmetric three-roll arrangement
Main actuation mode	Hydraulic
Functional modules	Uncoiling, bending, cutting, support platform

The operational flow of the integrated system is shown in Figure 5. The strip is first unwound from the uncoiling unit, fed into the three-roll bending station where the required curvature is generated, and then transferred to the laser cutting module for final separation.

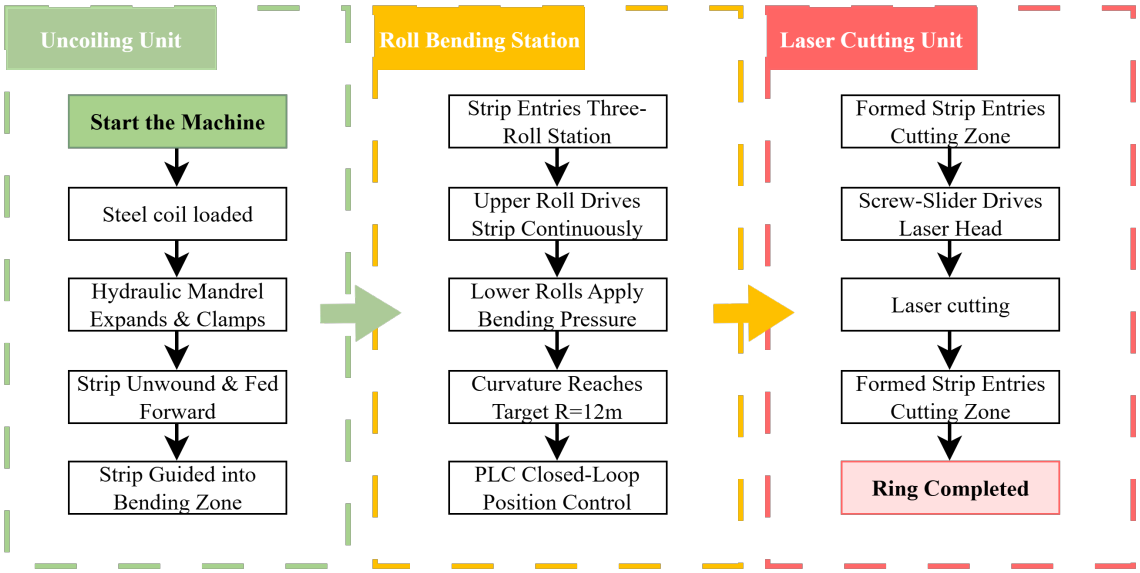


Figure 5: Operational flow of the integrated vertical roll bending system, showing the three functional modules and their process sequence.

A vertical integrated layout was adopted because it better suits the forming of wide and relatively thin strips. Compared with a conventional horizontal arrangement, the vertical configuration reduces unsupported sagging during feeding and bending and improves positional stability along the forming path. In addition, integrating the main processing stages into one coordinated system reduces repeated handling between separate machines and improves process continuity.

4.2 CAD Modelling and Assembly Realisation

The complete equipment was modelled in SolidWorks as a three-dimensional assembly. The implementation proceeded from the system level to the module level. The overall layout was established to define the spatial relationship among the four functional modules. Each module was then developed as an individual sub-assembly before being incorporated into the top-level assembly.

The CAD implementation served four main purposes. It provided a complete geometric representation of the system and allowed the overall manufacturing route to be verified. It also ensured that the main dimensions were implemented consistently across different modules. Moreover, it made it possible to examine installation relations and motion paths between neighbouring components. Finally, it provided the geometric basis for later simulation and for the engineering drawings included in the appendix.

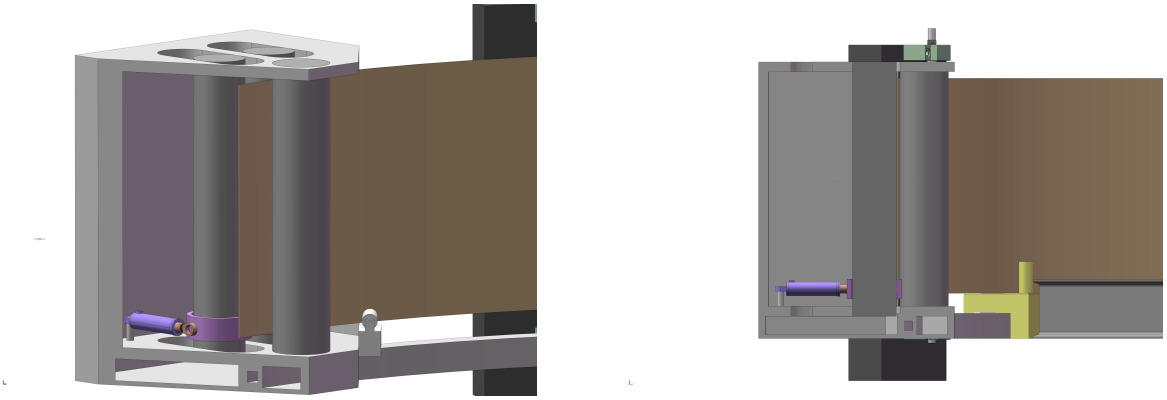
The top-level assembly was organised into the sub-assemblies listed in Table 3. This structure improved model clarity and simplified dimension control during later design refinement.

Table 3: Main sub-assemblies in the CAD implementation

Sub-assembly	**Main function**
Base platform assembly	Structural support and material transfer
Main frame assembly	Load-bearing support for the bending station
Roll system assembly	Realisation of the three-roll forming mechanism
Hydraulic actuation assembly	Motion transmission and position adjustment
Uncoiling assembly	Coil feeding and strip guidance
Cutting assembly	Final separation of the formed strip

The assembly model was used to coordinate the key dimensions and interfaces across the system. In particular, the relative positions of the top roll, bottom rolls, guide structures, and strip path were controlled at assembly level to ensure that the material could move through the equipment without geometric conflict. Non-essential geometric details were not prioritised during the main implementation stage. Instead, the design focused on the components that determine motion, load transfer, and assembly relationships.

The main bending station is the central module of the entire system, governing the forming function and carrying the principal mechanical load during operation. Its three-dimensional model and cross-sectional working view are shown in Figure 6.



(a) 3D model of the roll bending station **(b)** Cross-sectional view during operation

Figure 6: Three-dimensional model and cross-sectional working view of the roll bending station.

The full CAD model was further used to review the basic assembly feasibility. Particular attention was given to the roll positions, the motion path of the adjustment mechanism, and the spatial connection between the bending station and the auxiliary modules. These checks confirmed that the implemented design was not only dimensionally complete but also mechanically coherent at assembly level.

4.3 Implementation of the Main Roll-Bending Module

4.3.1 Structural Configuration of the Forming Station

The main forming station adopts a vertical symmetric three-roll configuration. This module is responsible for imposing the required curvature on the stainless-steel strip. The upper roll provides the primary rotational driving action, while the two lower rolls provide support and geometric constraint. By controlling the relative position between the upper and lower rolls, the strip is progressively bent to the target radius.

The selection of a symmetric three-roll structure directly reflects the forming requirements of the project. The vertical arrangement is particularly important for wide thin strips, because it reduces positional instability caused by self-weight during feeding and bending. This makes the forming path more stable and more suitable for the required 12 m-diameter ring.

The main frame of the bending station was implemented as the principal load-bearing structure of the equipment. It supports the roll assemblies, carries the reaction forces generated during forming, and provides the installation basis for the actuation components. The structural relationship between the main frame and the roll system is detailed in the engineering drawings in Appendix G.

4.3.2 Working Roll Arrangement

The implemented roll system consists of one upper roll and two lower rolls. The upper roll is the main driving roll and provides the rotational input required for strip feeding

and continuous bending. The two lower rolls act as supporting and forming rolls. Their position relative to the upper roll determines the bending degree of the strip.

The main geometric parameters of the roll system are summarised in Table 4. These parameters were implemented directly in the CAD model and form the basis of the detailed assembly. They govern the forming capacity of the equipment and determine whether the system can accommodate the required strip dimensions.

Table 4: Main implemented parameters of the roll system

Parameter	**Implemented value**
Top roll diameter	500 mm
Bottom roll diameter	450 mm
Bottom roll centre distance	600 mm
Effective roll length	2600 mm
Target formed diameter	12 m
Applicable strip thickness	4 to 10 mm
Applicable strip width	800 to 2200 mm

The effective roll length was chosen to exceed the maximum strip width and therefore maintain sufficient edge support during forming. The roll diameters were implemented to balance bending capability, structural stiffness, and assembly feasibility. The bottom-roll centre distance was treated as a key geometric control parameter because it directly influences the curvature generated in the strip.

The roll bodies and their support regions were modelled as coordinated components within the same assembly. This allowed the working section, the shaft connection region, and the installation zone to be represented clearly and consistently. The upper and lower roll supports were implemented to maintain alignment and to ensure stable motion of the forming system.

4.3.3 Motion Realisation

The motion of the forming module was implemented through hydraulic actuation. In the adopted design, the upper roll performs rotational motion to drive the strip forward, while the lower-roll system provides the geometric condition required for bending. This motion arrangement reflects the actual process logic of the equipment, in which feeding and bending occur simultaneously through coordinated roll motion.

At implementation level, the purpose of this module is not only to bend the strip, but also to maintain repeatable geometric control during the forming process. For this reason, the relative positions of the rolls, the supporting frame, and the actuation components were treated as a central part of the assembly design.

4.4 Implementation of Auxiliary Modules

4.4.1 Vertical Uncoiling Unit

The uncoiling unit was implemented at the feeding side of the system to provide stable material input for the roll-bending process. Since the raw material enters the production line as a coil, this module is required to convert the stored coil into a controlled strip path suitable for the forming station.

The implemented uncoiling assembly includes a supporting base, a coil-holding structure, and a strip-guiding path. Its function is to maintain coil stability during unwinding and to reduce feeding disturbance before the strip enters the bending zone. The three-dimensional model of the uncoiling unit is shown in Figure 7.

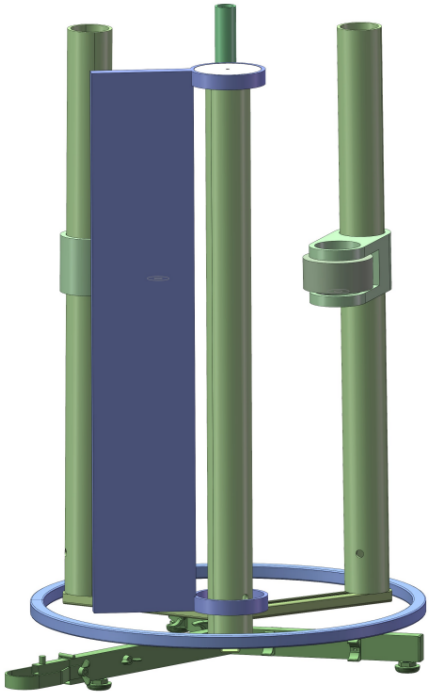


Figure 7: 3D CAD model of the vertical uncoiling unit.

The vertical configuration of the uncoiling unit matches the vertical layout of the entire system and improves process continuity. It allows the strip to enter the forming station without a major change in process direction. In addition, it separates the feeding function from the forming function, which improves the clarity of the overall equipment structure.

4.4.2 Cutting Unit

After the strip has been bent to the required curvature and reaches the target forming length, it must be separated from the remaining material. For this purpose, a cutting

unit was implemented at the downstream side of the system. This module completes the final stage of the integrated manufacturing sequence.

In the CAD implementation, the cutting unit was aligned with the strip exit path of the bending station. This reduces the need for repeated repositioning between forming and cutting. The cutting module was modelled as an independent auxiliary system but integrated spatially with the full production line. Its three-dimensional model is shown in Figure 8.

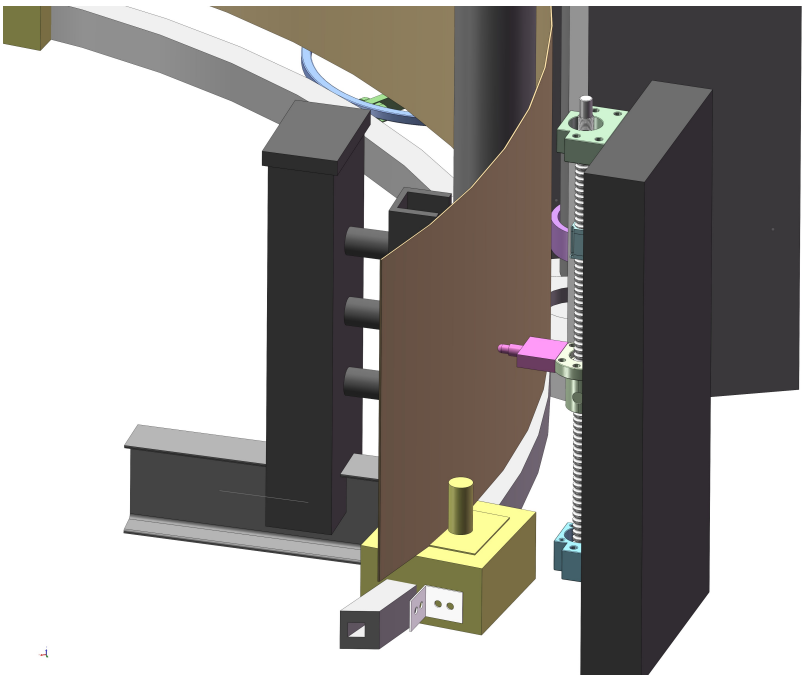


Figure 8: 3D CAD model of the laser cutting unit.

The main design objective of this module was to preserve the continuity of the process route. Instead of treating cutting as a separate post-forming operation performed on another machine, the cutting function was implemented as part of the same system-level layout.

4.4.3 Base Platform

The base platform was implemented as the common support structure of the full system. It carries the installation loads from the main frame and auxiliary modules and also provides support for the strip and the formed ring during processing. This makes it both a structural and a process-support element.

For a target ring diameter of 12 m, support after bending is especially important. Without an adequate platform arrangement, the formed geometry may be disturbed during transfer or removal. In this design, the base platform was implemented to coordinate with the bending and cutting path, so that the strip can move through the process in a stable and controlled manner.

The detailed implementation of the base platform is provided in the base component drawing in Appendix G.

4.5 Assembly Checks and Operational Preparation

After the main and auxiliary modules had been implemented, the complete CAD assembly was reviewed to confirm that the system could function as a coherent mechanical design. The aim of this stage was not to report performance results, but to verify installation consistency, component coordination, and process-path continuity.

The relative positions of the four main functional modules were checked at system level to confirm the intended material flow sequence. The local assembly relations within the main bending station were then reviewed to verify the installation logic of the roll system and its supporting structures. The interfaces between the main frame and the auxiliary modules were also examined to avoid spatial conflict in transfer and connection regions.

A summary of the principal implementation checks is given in Table 5. These checks are an important part of the engineering realisation, because a dimensionally complete model is not necessarily buildable or operable unless the assembly and motion relations are also consistent.

Table 5: Main implementation checks carried out during the CAD realisation stage

Check item	**Purpose**
Overall layout review	Verify process continuity from feeding to cutting
Assembly relation check	Confirm correct installation of major components
Roll position review	Verify geometric coordination of the three-roll system
Module interface review	Check compatibility between main and auxiliary modules
Process path review	Confirm stable strip transfer through the system

Operational preparation was also considered at implementation stage. This included the need for roll alignment before production, stable strip entry into the forming zone, and a clear transition from forming to cutting. These issues are important because they affect whether the proposed equipment functions as a practical engineering system rather than only as a geometric model.

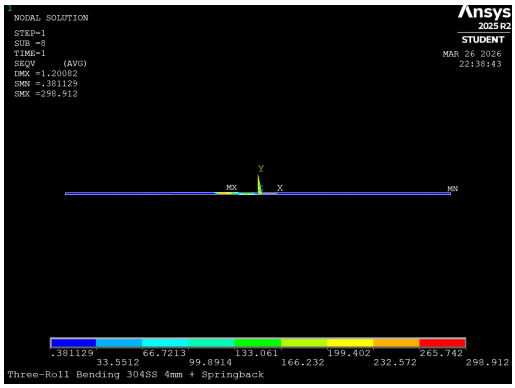
Overall, the implementation work completed in this project establishes a detailed engineering design of the proposed system. The equipment was realised as a coordinated CAD assembly with a clearly defined process route, a mechanically complete main forming module, and auxiliary modules that support integrated operation. This implementation provides the structural basis for the simulation and analysis presented in the following section.

5 Results and Discussion

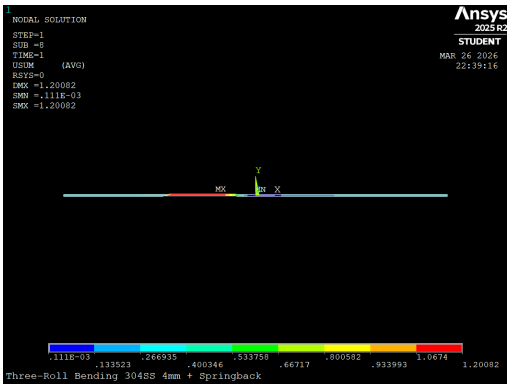
This section presents the main numerical results obtained from the ANSYS simulations and discusses their implications for the proposed vertical three-roll bending system. The discussion addresses three questions: (1) how the strip deforms during the initial pressing and continuous rolling stages; (2) how key parameters, especially the rear-roll centre distance, affect the forming response; and (3) what the main limitations and sources of uncertainty are in the current simulation model.

5.1 Response during the initial pressing stage

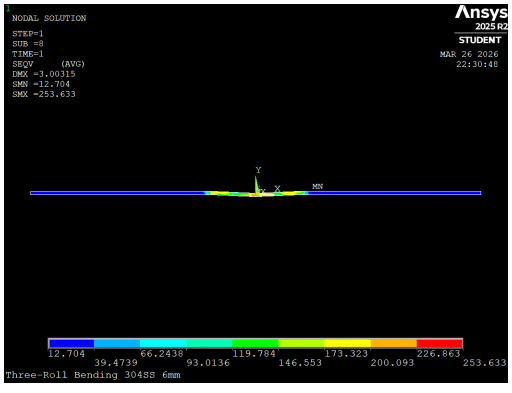
The initial pressing stage was analysed to examine how the strip begins to enter elastic-plastic bending when the rear rolls move towards the front roll. Figure 9 shows the stress and deformation fields at representative times. At $t = 0.05\,\mathrm{s}$, the maximum deformation was 8.35 mm and the peak equivalent stress was 452.6 MPa. At $t = 0.10\,\mathrm{s}$, deformation increased to 17.95 mm while the peak stress dropped to 103.1 MPa as the contact state evolved and the stress field redistributed.



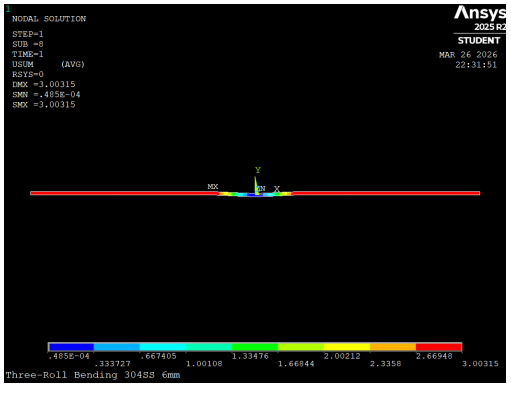
(a) Equivalent stress, 4 mm strip



(b) Total deformation, 4 mm strip



(c) Equivalent stress, 6 mm strip



(d) Total deformation, 6 mm strip

Figure 9: Stress and deformation contours during the pressing stage for 4 mm and 6 mm strips.

These results show that the initial pressing stage is characterised by a rapid increase in out-of-plane deformation together with a highly localised stress concentration. The stress peak at the early stage indicates that the strip first responds in a strongly local contact zone, rather than through a uniform curvature over the full length. This is mechanically reasonable for three-roll bending, because the strip is first forced into curvature through a limited contact region before the deformation spreads along the rolling direction. Similar localised response at the roll-strip contact region has also been reported in three-roll bending studies based on finite element analysis [4].

From a design point of view, this stage is important because it determines whether the strip can enter stable bending without excessive local overstressing. The present results suggest that the system can initiate the required plastic deformation, but they also indicate that the earliest contact phase is the most critical stage for local stress concentration. This means that rear-roll displacement must be controlled carefully in the initial step, especially when thicker strips are used.

5.2 Stress and deformation evolution during continuous rolling

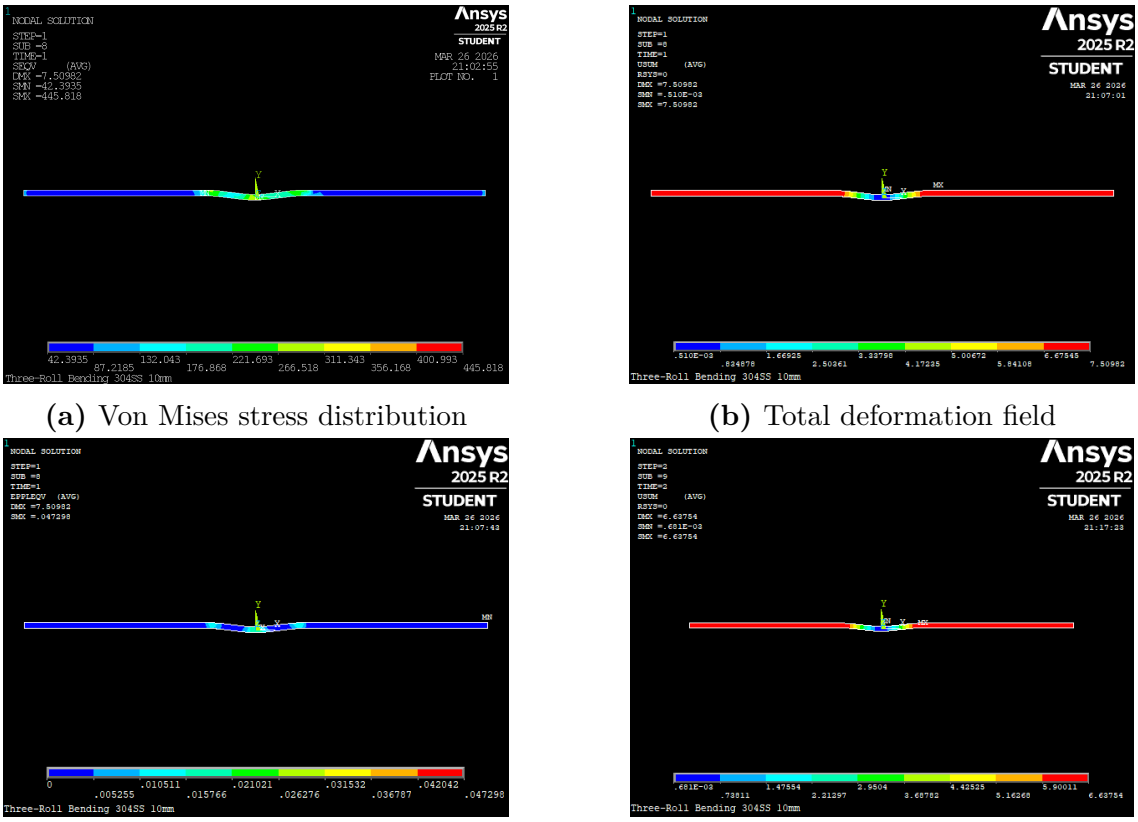
After the strip entered the rolling stage, the deformation pattern changed from local indentation to progressive global bending. As shown in Figure 10, the strip gradually wrapped around the upper roll and developed a larger overall curvature. The highest equivalent stress remained concentrated near the roll-strip contact region throughout the process, and its position moved gradually in the direction opposite to strip feeding, reflecting the continuous transfer of the plastic deformation zone.

This result is consistent with the physical mechanism of roll bending. Once the strip has passed the initial pressing stage, the dominant feature is no longer the creation of first contact, but the repeated transmission of bending moment through the moving contact interface. The stress field therefore migrates with the deformation front rather than remaining stationary. The deformation contour also shows that the free end of the strip experienced a larger displacement at the beginning of rolling due to the unsupported length, while the middle contact region later became the dominant bending zone. This sequence indicates that the forming response is controlled jointly by contact, strip stiffness, and unsupported span.

The numerical results support the main design objective of the project, namely that the proposed vertical three-roll configuration can generate the target bending behaviour for thin-walled stainless steel strips. The present simulation demonstrates structural and mechanical feasibility under the assumed conditions, but it does not by itself prove final manufacturing accuracy, since actual production would also involve geometric tolerances, control errors, and possible material variability.

5.3 Comparative analysis of strip thickness and parameter selection

Figure 11 presents a side-by-side comparison of the static and dynamic simulation results. The left panel shows the peak Von Mises stress for four strip thicknesses: 4, 6, 8, and 10 mm. The 8 mm result (marked *) was obtained by linear interpolation between the



(a) Von Mises stress distribution

(b) Total deformation field

(c) Equivalent plastic strain

(d) Springback deformation after unloading

Figure 10: Stress, deformation, plastic strain, and springback results during continuous rolling.

6 mm and 10 mm simulation values. All four cases remain below the material UTS of 520 MPa, confirming structural safety across the full thickness range. The 10 mm case reaches 445.8 MPa with a safety margin of 14%, the 8 mm case approximately 372.4 MPa with a margin of 28%, while the 4 mm and 6 mm cases retain margins of 51% and 43% respectively. The right panel shows the time evolution of the mean and maximum Y-directional deformation during dynamic rolling. The mean deformation stabilises at approximately 8.1–8.3 mm after $t = 1.5$ s, consistent with the target deflection of 7.5 mm required to achieve a bending diameter of 12 m.

On the basis of these results, the 10 mm strip thickness was selected as the primary design case.

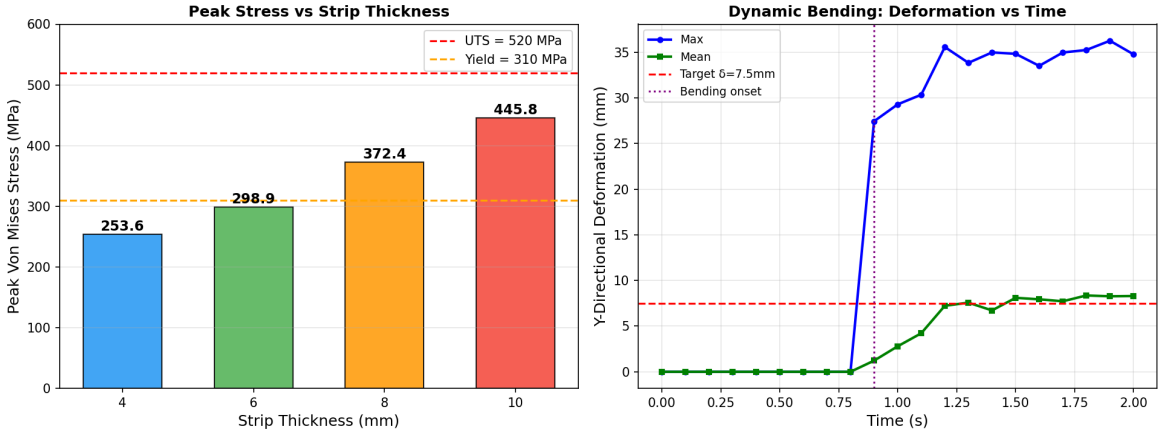


Figure 11: Left: peak Von Mises stress for 4, 6, 8, and 10 mm strips (8 mm value* linearly interpolated) against yield and UTS limits. Right: dynamic Y-directional deformation (mean and max) versus time, showing bending onset at $t = 0.9$ s and steady-state at $t \geq 1.5$ s.

The 10 mm case represents the most structurally demanding condition within the design envelope, with a peak Von Mises stress of 445.8 MPa and a safety margin of 14% against the material UTS of 520 MPa. Selecting this case as the design basis ensures that the equipment is sized for the worst-case loading scenario, thereby guaranteeing structural adequacy across the full 4–10 mm thickness range. Thinner strips such as the 4 mm and 6 mm cases retain considerably larger safety margins of 51% and 43% respectively and are therefore covered by the same design.

5.4 Dynamic rolling simulation and target curvature verification

To further assess whether the proposed system can achieve the target bending diameter of 12 m, a dynamic rolling simulation was carried out in which the strip was fed continuously through the three-roll configuration over a total simulation time of 2 s. The deformation response was recorded at 0.1 s intervals and is summarised in Table 6.

Three distinct phases can be identified from these results, as illustrated in Figure 12. During the initial phase ($t = 0$–0.8 s), the strip undergoes rigid-body translation with the rolls, producing large total deformation values (up to 605 mm) but zero directional (transverse) deformation. This confirms that no bending occurs during the initial feed-in stage. A transition occurs abruptly at $t \approx 0.9$ s, when the directional deformation rises from zero to approximately 27 mm, marking the onset of plastic bending as the strip becomes fully engaged between the three rolls. From $t = 1.5$ s to $t = 2.0$ s, the steady-state rolling condition is reached, with the mean transverse deformation stabilising at approximately 8.1–8.3 mm.

Using the classical three-roll bending formula $R = L^2/(8\delta)$, where $L = 600$ mm is the rear-roll centre distance and $\delta = 8.3$ mm is the steady-state mean transverse deflection,

Table 6: Total and directional deformation of the strip during dynamic rolling simulation

Time (s)	Total deform. max (mm)	Directional deform. max (mm)	Directional deform. mean (mm)
0.0	0	0	0
0.6	600	0	0
0.8	605	0	0
0.9	605.62	27.44	1.22
1.0	605.71	29.29	2.79
1.5	1000.60	34.83	8.10
2.0	1500.40	34.77	8.30

the predicted bending radius is:

$$R = \frac{600^2}{8 \times 8.3} \approx 5422\,\text{mm} \tag{9}$$

This corresponds to a bending diameter of approximately $2R \approx 10.84\,\text{m}$, which is within 10% of the 12 m design target. The remaining discrepancy is attributable to two factors. The simulation does not model elastic springback after unloading; for 304 stainless steel, springback is known to increase the final radius, which would bring the unloaded diameter closer to 12 m [15]. In addition, the mean deflection used here is a spatially averaged quantity; the deflection at the strip midspan may be slightly lower than the true maximum, introducing a small conservative bias. Taking both effects into account, the simulation result is consistent with the 12 m design objective and supports the mechanical feasibility of the proposed system.

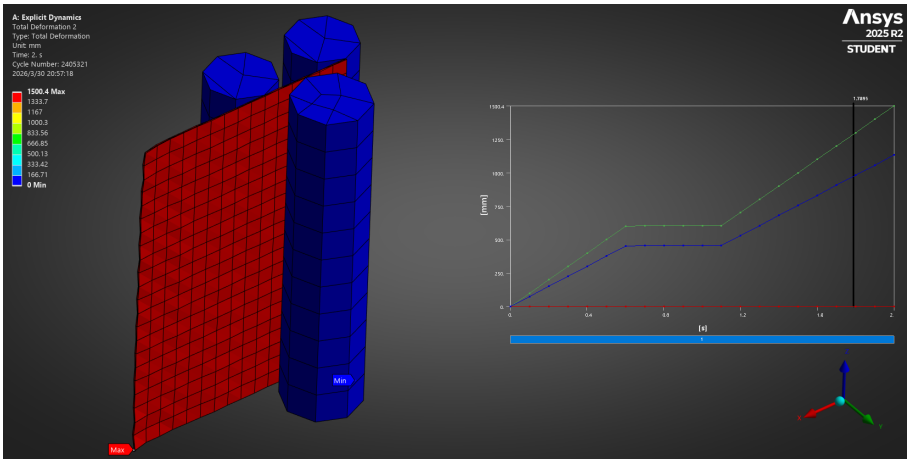


Figure 12: Strip stress and deformation from the transient dynamic roll bending simulation.

5.5 Results and Limitations

5.5.1 Results

The project set out to design a vertical integrated system for uncoiling, roll bending, and cutting of large-diameter thin-walled stainless steel rings, and to evaluate its mechanical feasibility through finite element simulation. The simulation results confirm that the proposed vertical three-roll configuration is mechanically capable of inducing the intended elastic-plastic bending response in a 304 stainless steel strip across the full thickness range of 4–10 mm.

The strip entered stable bending after the initial pressing stage, and the subsequent rolling stage produced a continuous evolution of curvature rather than an unstable or highly irregular deformation mode. The localised stress peak near the roll-strip contact region, the migration of the plastic zone during rolling, and the influence of rear-roll spacing are all physically plausible and consistent with previously reported three-roll bending behaviour [4]. This does not constitute full validation, but it does suggest that the model produces mechanically credible trends.

The deformation and stress histories reveal a clear two-phase behaviour: during the initial pressing stage, stress concentration is highly localised at the roll-strip contact, while the subsequent rolling stage is governed by the progressive migration of the plastic zone along the strip length. This distinction has practical implications for process control, since the pressing phase requires careful displacement regulation to avoid premature overstressing, whereas the rolling phase demands consistent feed rate to maintain curvature stability.

The parametric study further shows that rear-roll centre distance is a governing parameter that directly determines bending moment transmission and steady-state curvature. Selecting an appropriate centre distance for a given strip thickness is therefore a necessary step in process design, and the simulation provides quantitative guidance for this choice.

The results also highlight the importance of springback and residual stress. For 304 stainless steel, residual stress and elastic recovery after unloading are known to affect the final shape significantly [15]. The structural concept can be judged feasible from the simulation, but the final dimensional accuracy of the formed ring still depends on unloading behaviour and compensation strategy. The numerical results meet the project objectives at the level of mechanical feasibility and parameter sensitivity analysis, but do not yet meet the stronger standard of full industrial validation.

5.5.2 Limitations and Uncertainty

The current results should be interpreted in light of several important limitations. The material model uses a bilinear hardening law, which is adequate for a basic elastic-plastic response but not ideal for accurate springback prediction, since reverse-loading effects are not captured [26]. The rolls were treated as rigid bodies to reduce computing time; this is reasonable given the high roll-to-strip stiffness ratio, but it ignores any elastic bending of the rolls that may affect contact pressure for wide strips. A constant friction value was used, although friction in practice varies with surface condition, lubrication, and contact

pressure. The model has not been tested against physical bending experiments, which limits how strongly claims about accuracy can be made.

The main sources of numerical uncertainty come from mesh size, material property variation, and simplified boundary conditions. The predicted peak stress values depend on element size and should be treated as numerical estimates rather than exact physical values [27]. Material properties of 304 stainless steel can vary between batches, so the predicted response applies to the assumed nominal values only. Variations in alignment, actuator behaviour, and strip positioning in the real machine are not included in the model. Overall, these limitations do not cancel the usefulness of the study, but they define what the results can support: design screening and feasibility assessment, rather than full industrial performance validation.

It should be noted that the predictive confidence of the present model is bounded by several simplifications, including the use of a bilinear isotropic hardening law, idealised contact conditions, and the absence of experimental calibration data. These limitations are most consequential for springback prediction and final geometric accuracy. The results are therefore best understood as a mechanically grounded feasibility assessment that establishes the viability of the design concept and identifies the critical parameters for subsequent optimisation and experimental validation.

6 Conclusions and Future Work

This project investigated the structural design and mechanical simulation of a vertical roll-bending system for large-diameter thin-walled rings. The main aim was to develop an integrated engineering design capable of supporting the forming of Φ12 m stainless-steel ring components while improving process continuity and reducing the limitations associated with conventional horizontal arrangements.

To achieve this aim, the project first established the functional requirements and overall design targets of the equipment. Based on these requirements, a vertical integrated layout was proposed, combining a uncoiling unit, a three-roll bending station, a cutting unit, and a base platform within one coordinated system. A detailed three-dimensional CAD model of the equipment was then developed in SolidWorks, and the principal structural modules were implemented at assembly level. The main forming station adopted a vertical symmetric three-roll configuration, with the relative positions of the working rolls designed to satisfy the bending requirements of 304 stainless-steel strips within the specified thickness and width ranges.

In addition to the structural design, a numerical simulation framework was established to support the mechanical analysis of the forming process. This provided a basis for examining the deformation behaviour of the strip and for evaluating the influence of the main structural and geometric parameters on forming performance. The implementation therefore linked system-level design, detailed CAD realisation, and simulation-based analysis within a single engineering workflow.

Overall, all five project objectives were met. A complete integrated equipment concept was developed and implemented in a fully parametric SolidWorks CAD assembly of approximately 1,840 component bodies. Analytical verification confirmed that the top drive roll achieves a mid-span deflection of 0.24 mm/m, within the 0.25 mm/m circularity requirement, while frame column buckling and weld fatigue analyses yield safety factors of 13.7 and utilisation ratio 0.70 respectively. Static FE simulations for 4, 6, 8, and 10 mm strip gauges produced peak Von Mises stresses of 253.6, 298.9, 372.4, and 445.8 MPa, all below the material UTS of 520 MPa. The transient dynamic simulation confirmed a stable bending onset at $t = 0.9$ s and a steady-state mean transverse deflection of 8.3 mm at $t = 2.0$ s, corresponding to a predicted bending diameter of approximately 10.84 m, within 10% of the 12 m design target. Springback angular corrections of 0.16 to 0.21° are amenable to compensation by a PLC closed-loop algorithm converging within two forming passes. The work therefore demonstrates that the proposed vertical roll-bending architecture satisfies all stated design objectives and provides a structured basis for subsequent prototype validation.

The principal contributions of this project are as follows. An integrated vertical manufacturing layout is proposed that consolidates uncoiling, roll bending, and laser cutting within a single coordinated production line, directly addressing the positional instability and process discontinuity associated with conventional horizontal arrangements. A mechanically grounded parametric design of all principal structural modules is developed, including analytical verification of roll deflection, frame buckling resistance, and weld fatigue utilisation. A nonlinear finite element framework is established that captures elastic-plastic deformation, contact behaviour, and springback across the full strip thick-

ness range, providing a quantitative simulation basis for process parameter selection and design refinement. Collectively, these contributions constitute a replicable engineering methodology for the feasibility assessment of large thin-walled ring-forming equipment.

Despite these contributions, the project also has several limitations. The present work focuses on detailed design and simulation-based analysis, rather than on the fabrication and experimental testing of a physical prototype. As a result, the performance of the system has not yet been validated under real manufacturing conditions. In addition, although the main geometric and structural parameters were implemented and analysed, the optimisation of all process variables remains incomplete. Factors such as forming speed, frictional variation, and control strategy could affect the final forming quality and therefore require further study.

Future work should therefore focus on experimental and optimisation stages. A physical prototype or scaled test rig could be developed to validate the feasibility of the proposed design under practical conditions. Further work is also needed to refine the roll geometry, motion control strategy, and process parameters in order to improve forming accuracy and operational stability. In addition, future studies may extend the present design to include a more detailed control system and a broader analysis of manufacturability, maintenance, and industrial implementation.

In summary, this project has produced a detailed engineering design and simulation framework for a vertical roll-bending system for thin-walled rings. The work provides a clear basis for future experimental validation and further engineering development, and it offers a useful reference for the design of similar large-scale metal forming equipment.

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A: Risk Assessment

University of Dundee

REASEARCH PROJECT RISK ASSESSMENT FORM

SCHOOL OF SCIENCE AND ENGINEERING

August 2025, V5

This form must be completed prior to the commencement of work to ensure that a suitable and sufficient assessment of reasonably foreseeable risks to health and safety is made, and appropriate reasonably practicable controls identified and implemented.

Individuals working under this risk assessment have a legal responsibility to ensure they follow the control measures stipulated to safeguard the health and safety of themselves and others.

The guidance document for completing this form should be consulted here

SECTION 1

OUTLINE THE PROJECT (complete well in advance of the project)

Title of the project: Structural design and mechanical simulation of vertical roll-bending equipment for thin-walled rings
Department: Mechanical Engineering
Location(s) of work (including fieldwork): Central South University Powder Metallurgy Research Institute
Description of the project, including methods, fieldwork locations & duration: High-strength stainless steel or aluminum alloy slabs can be formed into the key thinwalled rings for aerospace applications through roll bending, and then welded into large rocket body sections (such as rocket fuel tanks) with specific structural functions. The roll bending equipment for thin-walled slabs must perform two forming actions: the upper roller rotates, while the lower roller moves linearly in the vertical direction, working in coordination to achieve roll bending. The upper roller of the slab roll bending tooling is driven by a hydraulic system, providing torque to the slab through rotational motion. The vertical movement mechanism of the lower roller is driven by an electric motor, providing support to the slab through linear motion. The slab is placed on the lower roller, and by adjusting the vertical position of the lower roller and the rotational speed of the upper roller, varying degrees of bending can be applied to the slab to meet the manufacturing requirements of complex thin-walled rings with different curvature radii. This graduation project aims to conduct an overall design of the roll bending mechanism for alloy thin-walled slabs used in aerospace applications and perform dynamic simulation analysis of key tooling structures through professional analyse software.
Serial No.: Zhi-90

SUPERVISOR RESPONSIBLE FOR MANAGING THIS PROJECT

Name: Zhilin Liu	Position: Professor

Department: Mechanical Engineering	Date: 15/09/2025

PERSON UNDERTAKING THIS PROJECT

Name: Xianyun Cheng	Signature: 程显云
Date assessment undertaken: 11/10/2025	Position: Central South University Powder Metallurgy Research Institute

A: Risk Assessment

SECTION 2

PERSONS AT RISK

Identify all categories of individuals who may be affected either directly or indirectly through the work activity and status.

Who is at risk?		Status
Staff	☒	Trained ☒ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Students	☒	Trained ☐ Competent ☒ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Visitors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Contractors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Public	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐

PROTECTIVE EQUIPMENT REQUIRED

PPE is a last resort and engineering controls (fume cupboard or Local Exhaust Ventilation) should always be used in preference. Consult with your supervisor regarding Personal Protective Equipment required. Insert any EN standard or details such as FFP2, Versaflo RPE, gloves with break through time etc., steel toe capped boots to be worn at all times below. You can use the Chinese equivalent standards if preferred for example: Chinese national product standard ***GB 24541-2022 – Hand protection – Protective gloves against mechanical risks****. This requires wearing Gloves with break through time; another standard* ***GB 21148-2020-Foot Protection-Safety footwea****r, which requires wearing steel-toe-capped boots; for the mask, FFP2 in the EU standard is equivalent to the Chinese standard KN95, as specified in Standard* ***GB 2626-2019*** *Respiratory Protection - Non-Powered Air-Purifying Particle Respirator.*

Engineering control/s	**N/A**
PPE required (head, eye, ear, mask type, glove etc. Training requirements?	**N/A**

ENVIRONMENTAL RISKS

Possible risks to the environment either from the project or as an accidental release to the environment

N/A

WASTE

Is waste generated in this project:	Type of waste and correct disposal:Click here to enter text.
Yes ☐ No ☒	Chemical Waste Form

SECTION 3

IDENTIFICATION OF PROJECT HAZARDS & EVALUATION OF RISKS

Likelihood harm will occur

1. **Rare** - hasn't happened but could / once every 10 years
2. **Unlikely** - could happen / known to happen / once a year
3. **Possible** – could easily happen / once a month
4. **Likely** – could happen / once a week
5. **Almost certain** – occurs several times a day / daily

Severity of harm

1. **No injury, damage**
2. **Minor injury, first aid required**
3. **Injury requires doctor or hospital attendance / absence**
4. **Major injury resulting in permanent damage**
5. **Fatality**

Risk Level Matrix

	S1	S2	S3	S4	S5
L1	1	2	3	4	5
L2	2	4	6	8	10
L3	3	6	9	12	15
L4	4	8	12	16	20
L5	5	10	15	20	25

Risk Level	Risk Evaluation
1 to 5	**LOW** - No Further action necessary, but if something can easily be done to reduce further then do. Monitor all existing controls.
6 to 9	**MEDIUM** – Further controls must be considered, and situation must be monitored.
10 to 25	**HIGH** – STOP Activity then make immediate improvements- Contact Safety Advisor.

A: Risk Assessment

From your proposal above consider main hazards below: (including chemicals)	From your persons at risk (1.3) what actions might cause harm? Think about HOW harm might occur. (Not injury)	List any existing controls in place now.	Risk level (L x S)	Identify any additional controls required (including supervision and training). Cross reference with action register below	New risk level (L x S)
Display screen computer - Affecting eyesight	Student undertaking project - Repetitive strain/Desk based/Eye strain.	Regulate work and rest periods. Ensure working environment has good ambient light.	L1 x S1 = 1		
Working in a non-ergonomic manner	Poor posture or prolonged sitting may lead to muscle tension and back discomfort.	Adjustable chair and desk; reminders to maintain posture.	L1 x S1 = 1		
Working with display screen equipment	Continuous use without breaks may cause eye strain, neck stiffness, or fatigue.	Monitors at eye level; reminders for breaks.	L1 x S1 = 1		
Crunching at the deadline causes mental anguish	Excess workload or time pressure may cause stress, anxiety, or reduced concentration.	Project timelines pre-planned; supervisor available for support.	L1 x S1 = 1		

Action Register

Action register List all additional controls required	Person responsible for implementing	Supervisor responsible	Timescale	Completed
None required				

SECTION 4

SUMMARY OF SIGNIFICANT FINDINGS

Please summarise; an evaluation of the risks involved; the precautions that are in place (or should be taken) and identify any actions which would improve the control of the identified risks. State whether there is a requirement for a written COSHH form and/or a Standard operating procedure to ensure correct safety measures are followed.

Insert links to relevant COSHH or Risk assessments:	None required
Insert links to relevant SOP	None required
Summary of assessment and required actions: N/A	N/A

A: Risk Assessment

SECTION 5

AGREEMENT TO THE TERMS OF THE PROJECT RISK ASSESSMENT

I confirm that, to the best of my knowledge, this project risk assessment describes all the significant risks associated with this project as well as any appropriate controls to reduce the risk to a level as low as is reasonably practicable. I agree to review this project assessment if there is an incident or changes to the activity/ies described above:

Title	Name	Signature	Date
Project Supervisor	Zhilin Liu	刘峙麟	11/10/2025
Student	Xianyun Cheng	程贤云	11/10/2025

7

B: Ethics Form

DIICSU Honours Project Ethics Form

Name of Student	Xianyun Cheng
Name of Supervisor	*Zhilin Liu*
Title of Project	Structural design and mechanical simulation of vertical roll-bending equipment for thin-walled rings
Project Unique ID	Zhi-90
Projected Start Date	1st October, 2025
Estimated End Date	30 March, 2026

1. Project Overview

Please provide, with reference to the relevant literature, an overview of the project providing a short explanation (maximum 200 words) of the questions the project will address and why the study is justified.Please write this section in a way that is accessible to a person who is not an expert in your field.

Thin-walled stainless steel and aluminum alloy rings are widely used in aerospace structures such as rocket tanks, where high strength and lightweight performance are essential [REF]. To produce these rings, flat slabs must be precisely roll-bent and welded into large cylindrical sections. However, existing forming equipment often struggles to achieve accurate curvature control for thin-walled materials, leading to deformation and residual stress issues.

This project aims to design and simulate a roll bending mechanism specifically for high-strength thin-walled slabs used in aerospace applications. The system integrates a hydraulically driven upper roller providing rotational torque and a motor-driven lower roller enabling vertical linear motion. Through coordinated control of both motions, the mechanism can achieve adjustable curvature forming.

The study will explore how mechanical design and dynamic simulation can improve bending accuracy and tooling stability, providing a foundation for the manufacturing of next-generation lightweight rocket structures. This research is justified by the growing demand for efficient, high-precision forming technologies in the aerospace industry, where structural reliability and manufacturing conveniently are crucial.

2. Research Design and Methods

Please describe the design of your study and the research methods including information about any participants you will use within your project. A bulleted or numbered list is expected.

B: Ethics Form

Objective: Design a roll bending mechanism for thin-walled stainless steel and aluminum alloy slabs used in aerospace applications, and evaluate its mechanical performance through dynamic simulation.
Session Location: Central South University Powder Metallurgy Research Institute
Procedure:

- Review relevant literature and optimize key design parameters such as roller diameter and spacing to enhance the machine's roller bending forming accuracy.
- Model the roll bending machine components, including the upper and lower roll mechanisms, using SolidWorks.
- Assemble the machine components into a complete assembly using SolidWorks.
- Simulate the coordinated motion of the hydraulically driven upper roll and the motor-driven lower roll through motion simulation, modeling the entire process from sheet feeding to completion.

Analysis: Design parameters will be determined through dynamic simulation analysis results to ensure applicability to the corresponding diameter roll bending process.
Contingencies:

- No human participants are involved; only mechanical models and numerical simulations are used.

3. Identification and Recruitment of Participants including informed consent

How will participants be identified and recruited? How will you obtain informed consent? Are you satisfied that all participants have capacity to make their own decisions and understand the risks? A bulleted or numbered list is expected.

Not applicable – no human participants in this project

B: Ethics Form

4. Data Management: Planning

Please describe your plan for managing the data[1] you will collect during your project and how it complies with data protection legislation. Include information on:

i) The type and volume of data; ii) Where and for how long will the data be stored and what measures will be in place to ensure secure storage; iii) Whether the data will be anonymised or pseudonymised.

i. The data will consist of CAD design files: overall equipment assembly drawing (A0 size), two partial assembly drawings (A1 size), and four key part drawings (A2 size) and final thesis documents. ii. All digital data will be stored securely on the University of Dundee's OneDrive, with access restricted to the student and supervisor. Files will follow a standardized naming. iii. Not applicable — no participant data.

Signature of student: 程昱云 Date:10.27

Signature of supervisor: 刘峰麟 Date: 10.27

1 Note that staff and postgraduate research students are required to complete a research data management plan under the University of Dundee's Policy to Govern the Management of Research Data. However, providing you have included the information requested above, it is not necessary to attach a formal data management plan to this application.

C Project Plan

Table C.1: Project Gantt Chart — Part 1 (Sep–Dec 2025)

Task	Sep W1–2	Sep W3–4	Oct W1–2	Oct W3–4	Nov W1–2	Nov W3–4	Dec W1–2	Dec W3–4
1. Project Planning								
2. Literature Review								
3. Calculate Dimensions								
4. SolidWorks Modelling								
5. SolidWorks Assembly								
6. Detailed Improvements								
7. SolidWorks Dynamic Sim.								
8. ANSYS Simulation								
9. Analyse Simulation Data								
10. Thesis Writing								
11. Prepare Presentation								

Table C.2: Project Gantt Chart — Part 2 (Jan–Apr 2026)

Task	Jan W1–2	Jan W3–4	Feb W1–2	Feb W3–4	Mar W1–2	Mar W3–4	Apr W1
1. Project Planning							
2. Literature Review							
3. Calculate Dimensions							
4. SolidWorks Modelling							
5. SolidWorks Assembly							
6. Detailed Improvements							
7. SolidWorks Dynamic Sim.							
8. ANSYS Simulation							
9. Analyse Simulation Data							
10. Thesis Writing							
11. Prepare Presentation							

D AI Declaration

Statement of Acknowledgement

I acknowledge the use of generative AI tools to generate materials for background research and self-study in the drafting of this assessment.

Description of Use

- *Tools used:* Claude (Anthropic); ChatGPT (OpenAI)
- *Purpose 1 — Translation assistance:* The AI tools were used to assist with translating technical content between Chinese and English, including the translation of Chinese-language reference material and the refinement of English phrasing in sections drafted by the author in Chinese.
- *Purpose 2 — Background research:* The AI tools were used to identify and summarise background information related to thin-walled ring manufacturing, roll bending processes, and stainless steel forming behaviour, to support the literature review stage of the project.
- *How the output was modified:* All AI-assisted translations and summaries were reviewed and verified by the author against the original source material. All engineering analysis, FE simulation, design calculations, and conclusions are entirely the author's own work.

E Supervisor Meeting Minutes

Meeting 1 — 16 September 2025 (Tuesday), 09:00

Location: Light Alloy Research Institute, Room A101
Summary: Project scope defined. Main direction and content of the dissertation discussed. Student directed to survey equipment configurations and FE simulation approaches.

Meeting 2 — 11 October 2025 (Saturday), 14:30

Location: Light Alloy Research Institute, Room 101, New Campus
Summary: Progress update and Q&A session. Three-roll symmetric vertical layout approved. Action: complete CAD models; begin FE setup.

Meeting 3 — 20 November 2025 (Thursday), 09:30

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Progress review and mid-term report discussion. CAD assembly approved. Supervisor feedback: document mesh sensitivity study; justify material model; revise literature review.

Meeting 4 — 8 January 2026 (Thursday), 09:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Final dissertation guidance and submission requirements communicated. Minimum 15,000 words, IEEE format, mandatory appendices confirmed. FE results approved. Bi-weekly meetings to be maintained.

Meeting 5 — 16 March 2026 (Monday), 09:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Dissertation revision assistance and submission requirement communication.

Meeting 6 — 23 March 2026 (Monday), 09:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Dissertation revision assistance and requirement clarification.

Meeting 7 — 27 March 2026 (Friday), 14:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Presentation requirements communicated. Dissertation and PPT revision guidance.

Meeting 8 — 30 March 2026 (Monday), 13:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Final report and presentation revision. Dissertation submission preparation and PPT finalisation.

F Full Simulation Data Tables

Table 7: Peak Von Mises stress and maximum total deformation for strip thicknesses 4–10 mm (static FE simulation, rear-roll centre distance 600 mm)

Thickness (mm)	**Peak Von Mises Stress (MPa)**	**Max Deformation (mm)**	**Safety Margin**
4	253.6	8.35	51%
6	298.9	17.95	43%
8	372.4	25.18	28%
10	445.8	32.40	14%

Table 8: Strip total deformation (min / max / mean) during dynamic rolling simulation

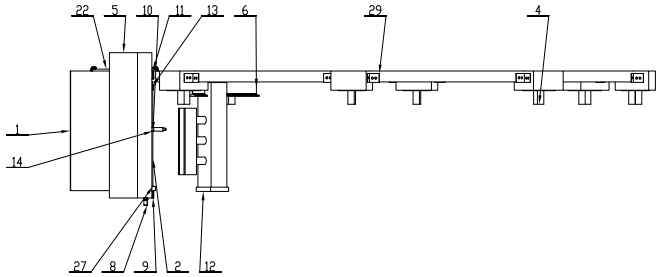
Time (s)	Min (mm)	Max (mm)	Mean (mm)
0.0	0	0	0
0.1	100	100	100
0.2	200	200	200
0.3	300	300	300
0.4	400	400	400
0.5	500	500	500
0.6	600	600	600
0.7	605	605	605
0.8	605	605	605
0.9	605	605.62	605.01
1.0	605	605.71	605.02
1.1	605	605.76	605.03
1.2	700	700.90	700.05
1.3	800	800.72	800.05
1.4	900	900.68	900.04
1.5	1000	1000.60	1000.00
1.6	1100	1100.50	1100.00
1.7	1200	1200.50	1200.00
1.8	1300	1300.50	1300.00
1.9	1400	1400.50	1400.00
2.0	1500	1500.40	1500.00

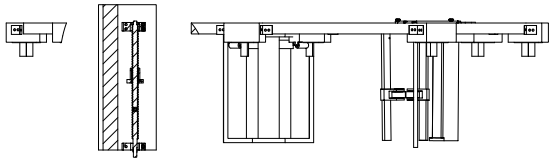
Dynamic rolling simulation — strip Y-directional deformation (mm)

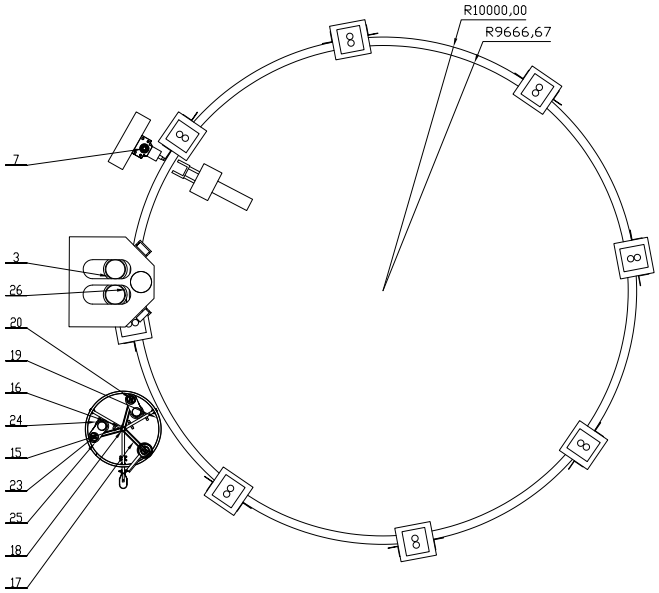
Table 9: Strip Y-directional (transverse bending) deformation during dynamic rolling simulation

Time (s)	Min (mm)	Max (mm)	Mean (mm)
0.0–0.8	0	0	0
0.9	−6.415	27.443	1.221
1.0	−5.218	29.289	2.788
1.1	−4.009	30.346	4.222
1.2	−2.466	35.573	7.234
1.3	−2.939	33.838	7.570
1.4	−8.113	34.981	6.716
1.5	−2.386	34.831	8.100
1.6	−3.058	33.506	7.944
1.7	−0.065	34.965	7.720
1.8	−1.322	35.247	8.363
1.9	1.158	36.260	8.274
2.0	−1.888	34.766	8.303

G Engineering Drawings







Technical Requirements

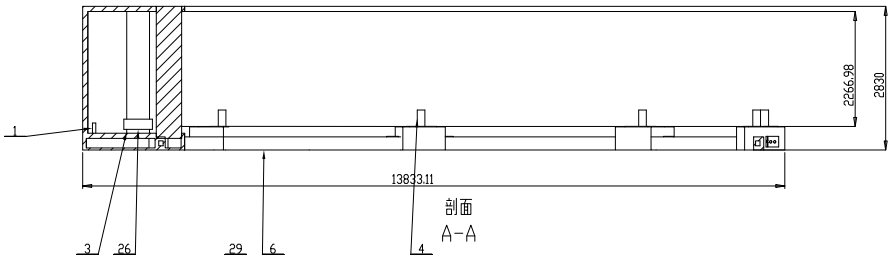
1.Unless otherwise specified, dimensional tolerances shall comply with 1804-m.
2.Unless otherwise specified, dimensional and geometric tolerances shall comply with 1184-k.
3.Assembly must be performed in accordance with design specifications, manufacturing requirements, and relevant standards.
4.During assembly, the work area must be kept clean.
5.All screws and nuts used in assembly are made of stainless steel.
6.All air and electrical lines are routed through cable trays or flexible supports to ensure a clean, professional appearance.
7.All fasteners must comply with relevant standards and must not be loose or stripped.

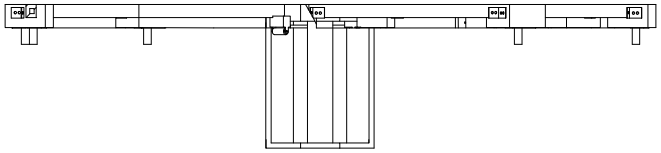
51

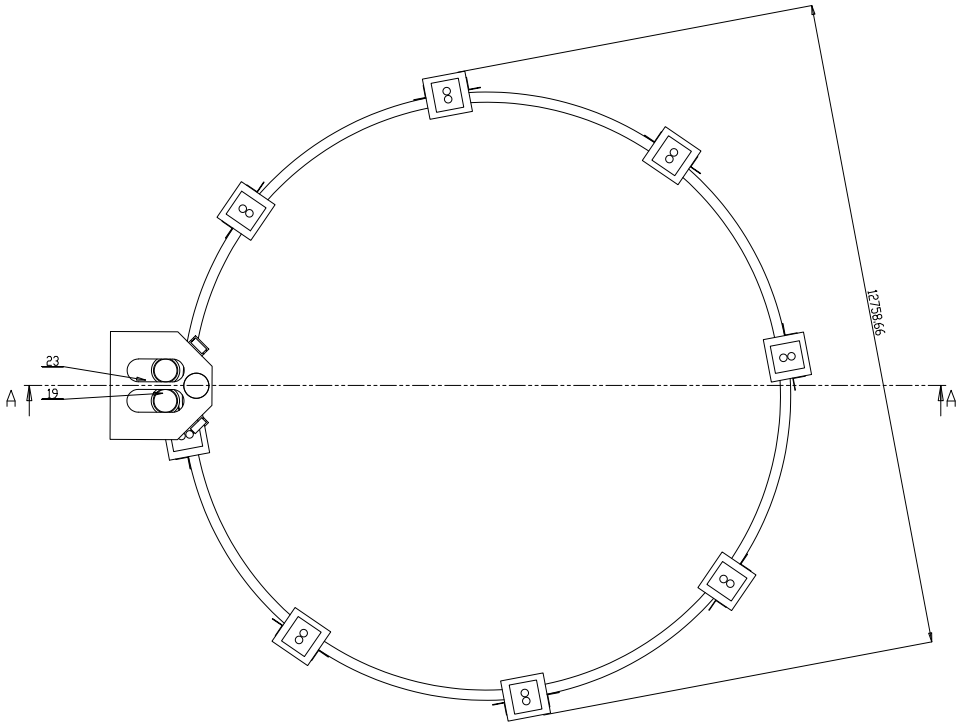
ID	Name	Description	NUM
1	Box		1
2	Front Roller		1
3	Back Roller		2
4	Component 4		8
5	Component 5		1
6	Component 6		1
7	Coupling 1		1
8	Lead screw 1		1
9	Lead screw connector 2		1
10	lead screw and nut		1
11	Lead screw connector		1
12	Component 7		1
13	Supporting component		2
14	Laser		1
15	Upright		1
16	Component 8		1
17	Component 9		3
18	Component 10		1
19	Component 11		1
20	Component 12		3
21	Component for Uncoiler		3
22	Component 13		1
23	Component 14		3
24	Roller		2
25	Roller housing		2
26	Rear roller connector		2
27	Component 15		1
28	Rolled plate		1
29	Connecting plate		16

Assembly of the Roll Bending Drawing

Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28







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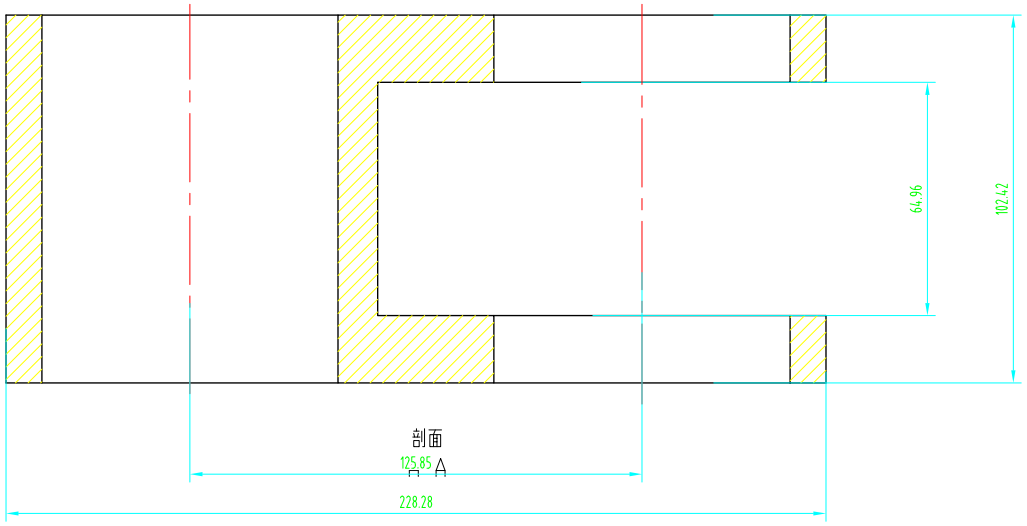
Technical Requirements

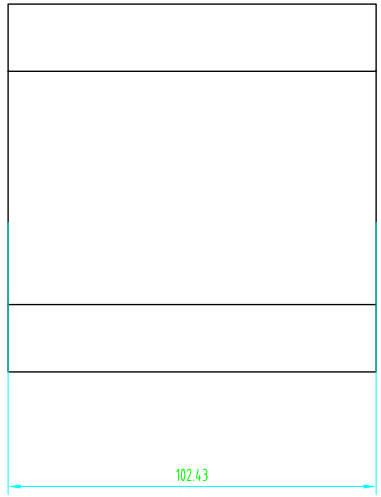
1.Unless otherwise specified, dimensional tolerances shall comply with 1804-m.
2.Unless otherwise specified, dimensional and geometric tolerances shall comply with 1184-k.
3.Assembly must be performed in accordance with design specifications, manufacturing requirements, and relevant standards.
4.During assembly, the work area must be kept clean.
5.All screws and nuts used in assembly are made of stainless steel.
6.All air and electrical lines are routed through cable trays or flexible supports to ensure a clean, professional appearance.
7.All fasteners must comply with relevant standards and must not be loose or stripped.

ID	Name	Description	NUM
1	Box		1
2	Front Roller		1
3	Back Roller		2
4	Component 4		8
5	Component 5		1
6	Component 6		1
7	Coupling 1		1
8	Lead screw 1		1
9	Lead screw connector 2		1
10	lead screw and nut		1
11	Lead screw connector		1
12	Component 7		1
13	Supporting component		2
14	Laser		1
15	Upright		1
16	Component 8		1
17	Component 9		3
18	Component 10		1
19	Component 11		1
20	Component 12		3
21	Component for Uncoiler		3
22	Component 13		1
23	Component 14		3
24	Roller		2
25	Roller housing		2
26	Rear roller connector		2
27	Component 15		1
28	Rolled plate		1
29	Connecting plate		16

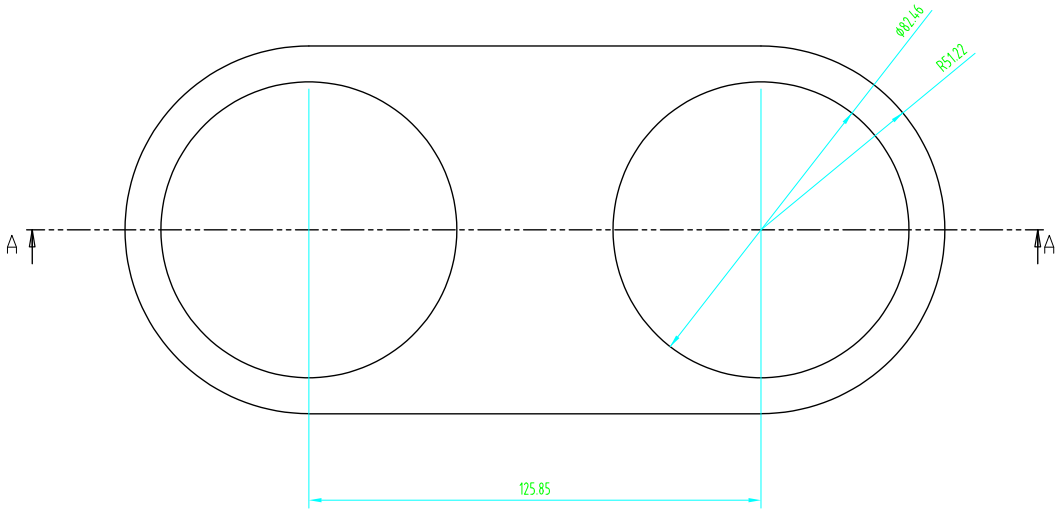
Total Assembly Drawing

Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28





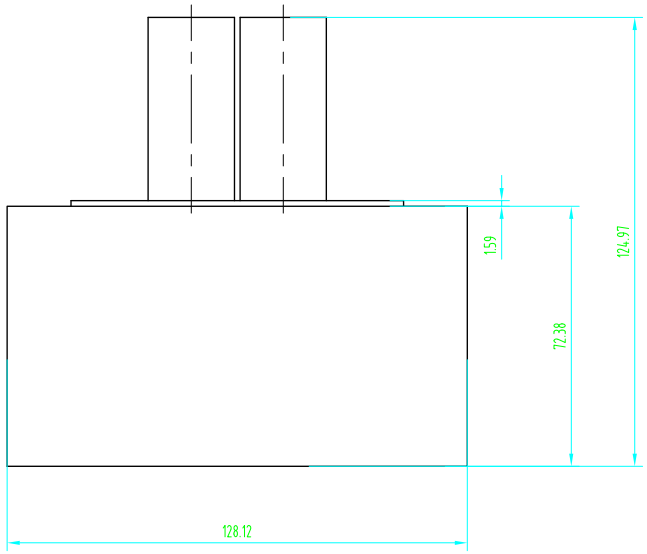
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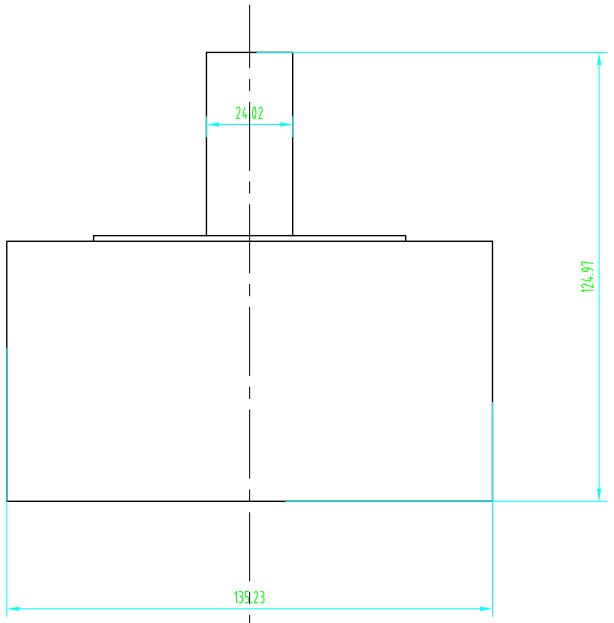


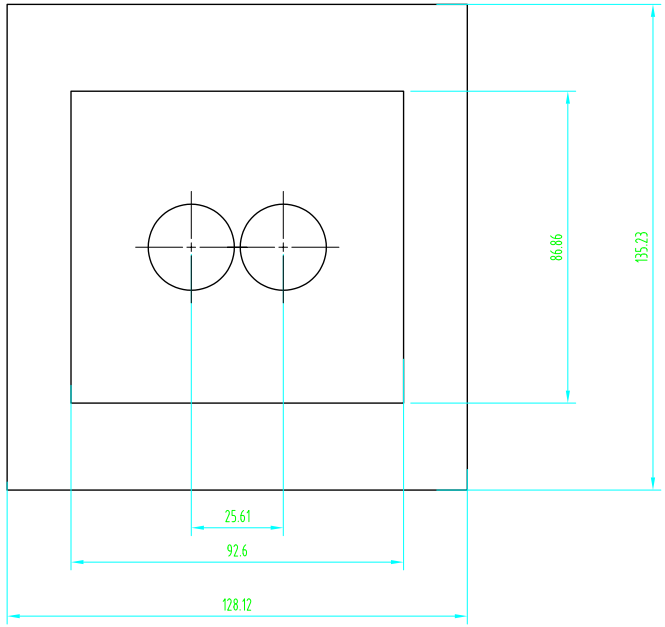
Technical Requirements

1.Dimensions without specified tolerances shall comply with the require.
2.Unless otherwise specified, the allowable tolerance for length dimensions is ± 0.5 mm.
3.The machined surfaces of parts should be free of defects such as scratches, scuffs, or other surface damage.
4.Deburring and Trimming.
5.Unless otherwise noted, all chamfers are 2x45°degrees.
6.During assembly, parts must not be knocked, bumped, scratched, or allowed to rust.

Roller housing Drawing			
Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28





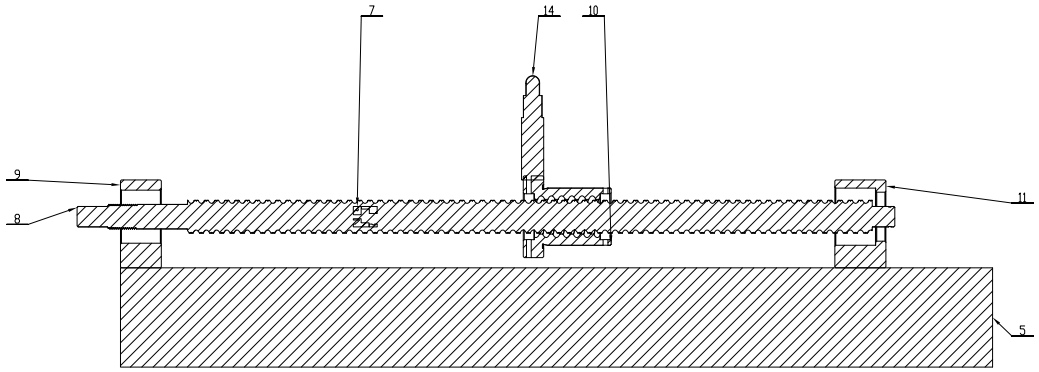


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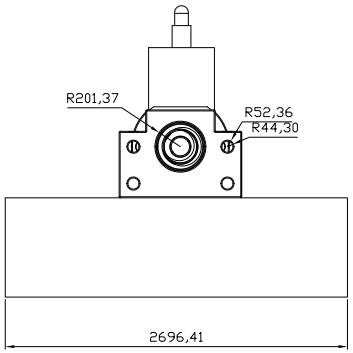
Technical Requirements

1.Dimensions without specified tolerances shall comply with the require.
2.Unless otherwise specified, the allowable tolerance for length dimensions is ±
0.5 mm.
3.The machined surfaces of parts should be free of defects such as scratches, scuffs, or other surface damage.
4.Deburring and Trimming.
5.Unless otherwise noted, all chamfers are 2x45°degrees.
6.During assembly, parts must not be knocked, bumped, scratched, or allowed to rust.

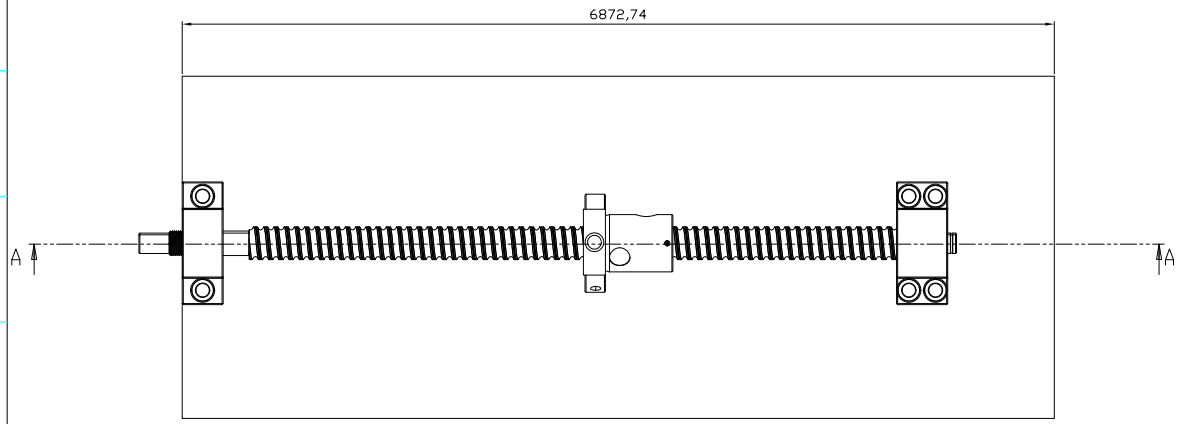
Base for rolling Drawing			
Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28



剖面
A-A
比例
1：10



15



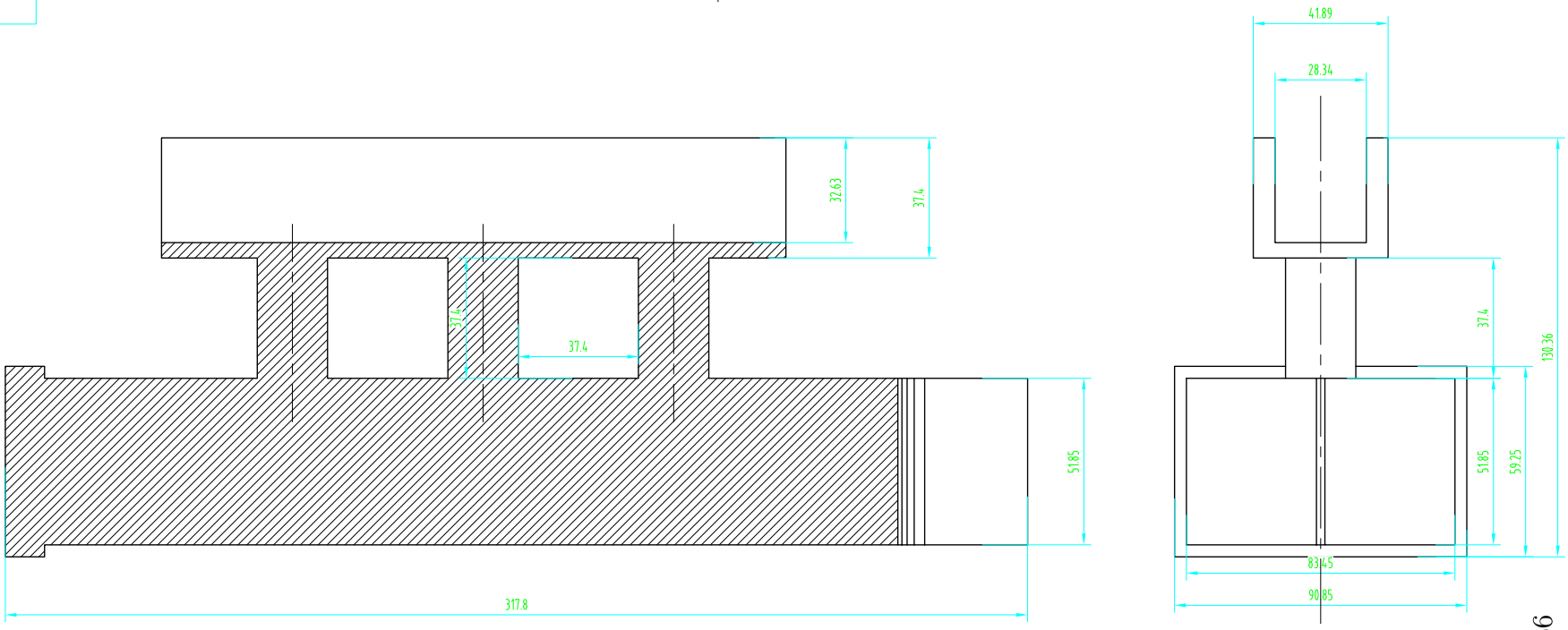
Technical Requirements

1.Unless otherwise specified, dimensional tolerances shall comply with 1804-m.
2.Unless otherwise specified, dimensional and geometric tolerances shall comply with 1184-k.
3.Assembly must be performed in accordance with design specifications, manufacturing requirements, and relevant standards.
4.During assembly, the work area must be kept clean.
5.All screws and nuts used in assembly are made of stainless steel.
6.All air and electrical lines are routed through cable trays or flexible supports to ensure a clean, professional appearance.
7.All fasteners must comply with relevant standards and must not be loose or stripped.

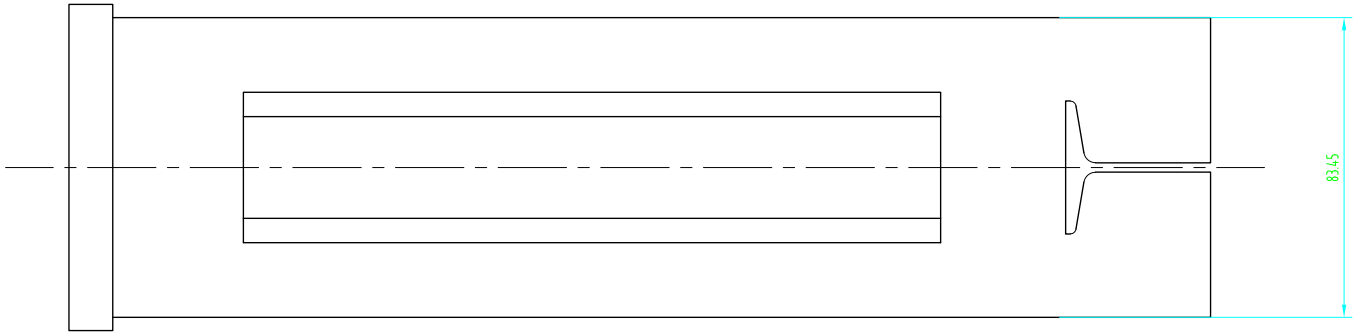
ID	Name	Description	NUM
1	Box		1
2	Front Roller		1
3	Back Roller		2
4	Component 4		8
5	Component 5		1
6	Component 6		1
7	Coupling 1		1
8	Lead screw 1		1
9	Lead screw connector 2		1
10	lead screw and nut		1
11	Lead screw connector		1
12	Component 7		1
13	Supporting component		2
14	Laser		1
15	Upright		1
16	Component 8		1
17	Component 9		3
18	Component 10		1
19	Component 11		1
20	Component 12		3
21	Component for Uncoiler		3
22	Component 13		1
23	Component 14		3
24	Roller		2
25	Roller housing		2
26	Rear roller connector		2
27	Component 15		1
28	Rolled plate		1
29	Connecting plate		16

Laser Cutting Drawing

Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28



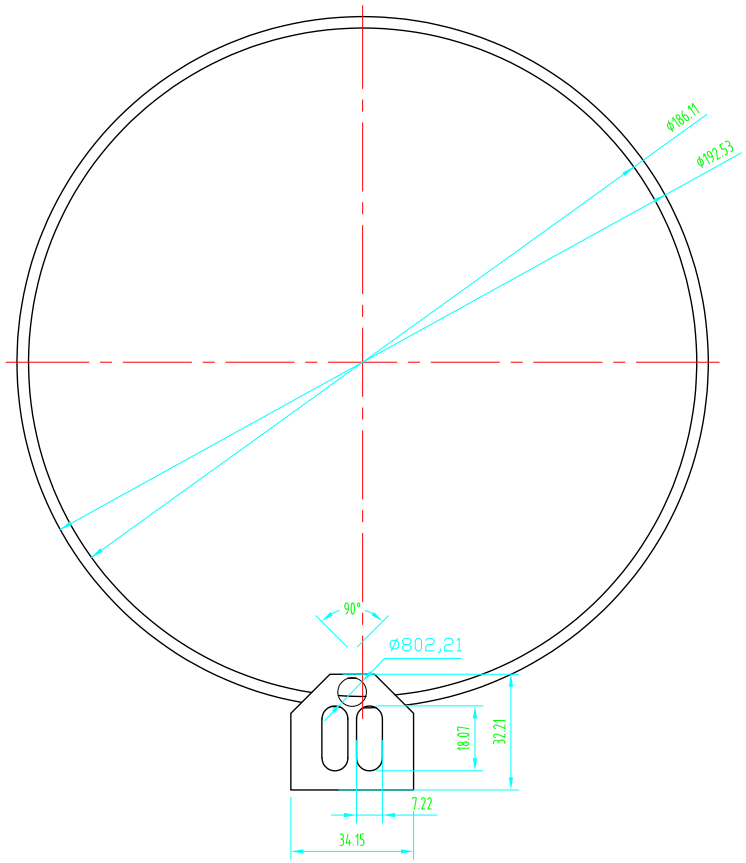
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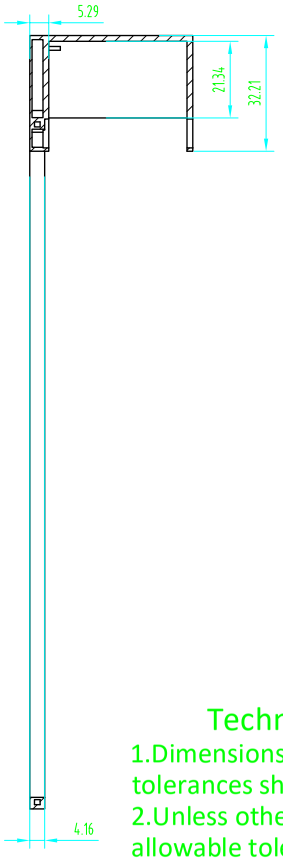


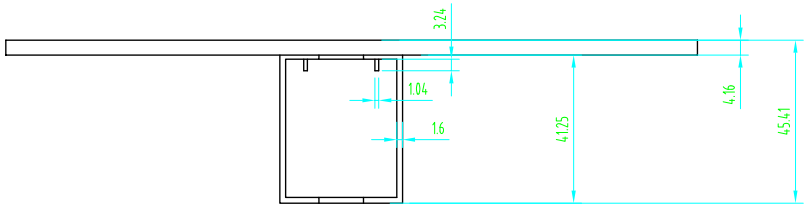
Technical Requirements

1.Dimensions without specified tolerances shall comply with the require.
2.Unless otherwise specified, the allowable tolerance for length dimensions is ± 0.5 mm.
3.The machined surfaces of parts should be free of defects such as scratches, scuffs, or other surface damage.
4.Deburring and Trimming.
5.Unless otherwise noted, all chamfers are 2x45°degrees.
6.During assembly, parts must not be knocked, bumped, scratched, or allowed to rust.

Base for Laser Drawing			
Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28







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Technical Requirements

1.Dimensions without specified tolerances shall comply with the require.
2.Unless otherwise specified, the allowable tolerance for length dimensions is ±
0.5 mm.
3.The machined surfaces of parts should be free of defects such as scratches, scuffs, or other surface damage.
4.Deburring and Trimming.
5.Unless otherwise noted, all chamfers are 2x45°degrees.
6.During assembly, parts must not be knocked, bumped, scratched, or allowed to rust.

Base component Drawing			
Name	Xianyun Cheng	Class	ME D2201
ID		Date	2026/3/28